Machine Learning CS 4641-7641

Decision Tree

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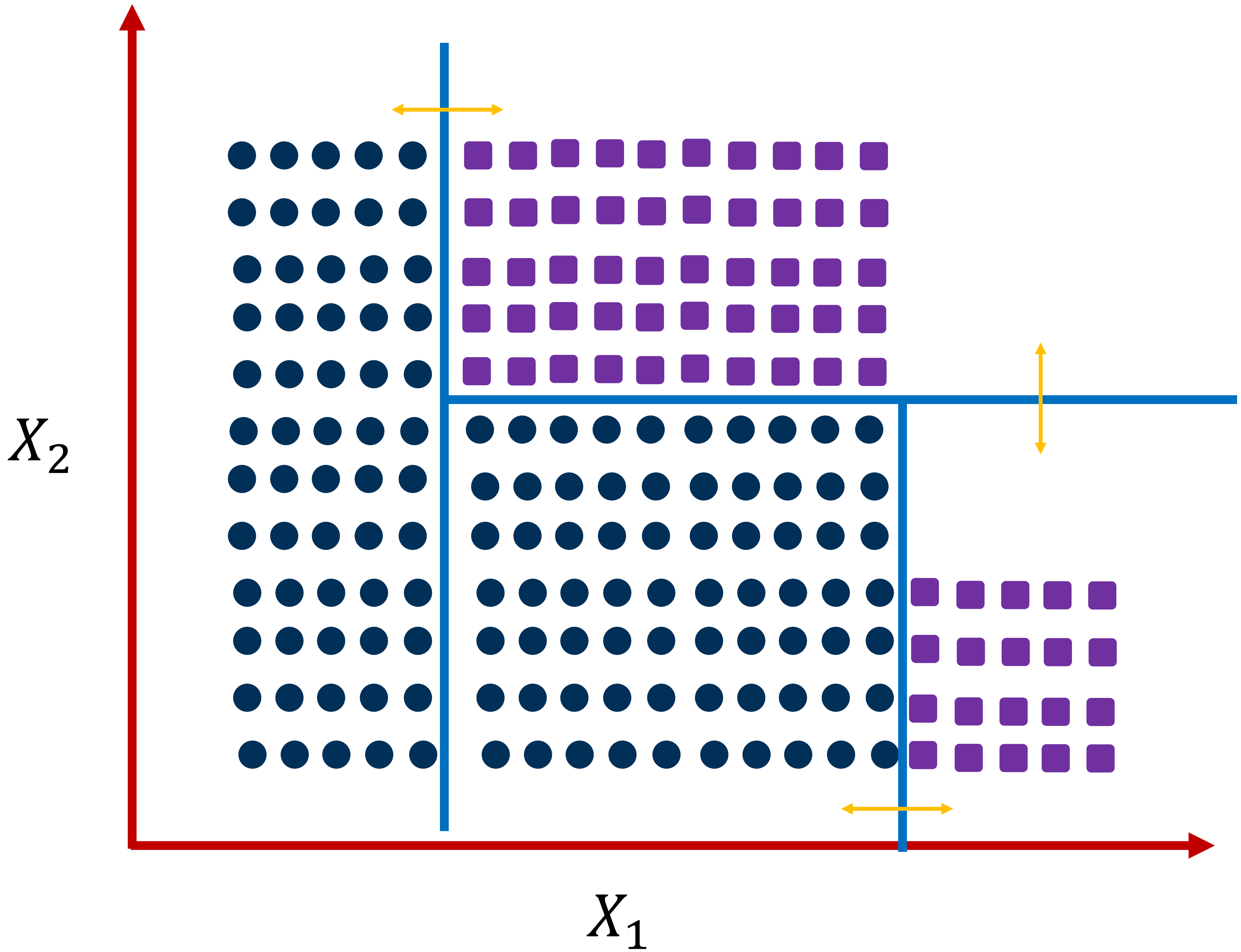
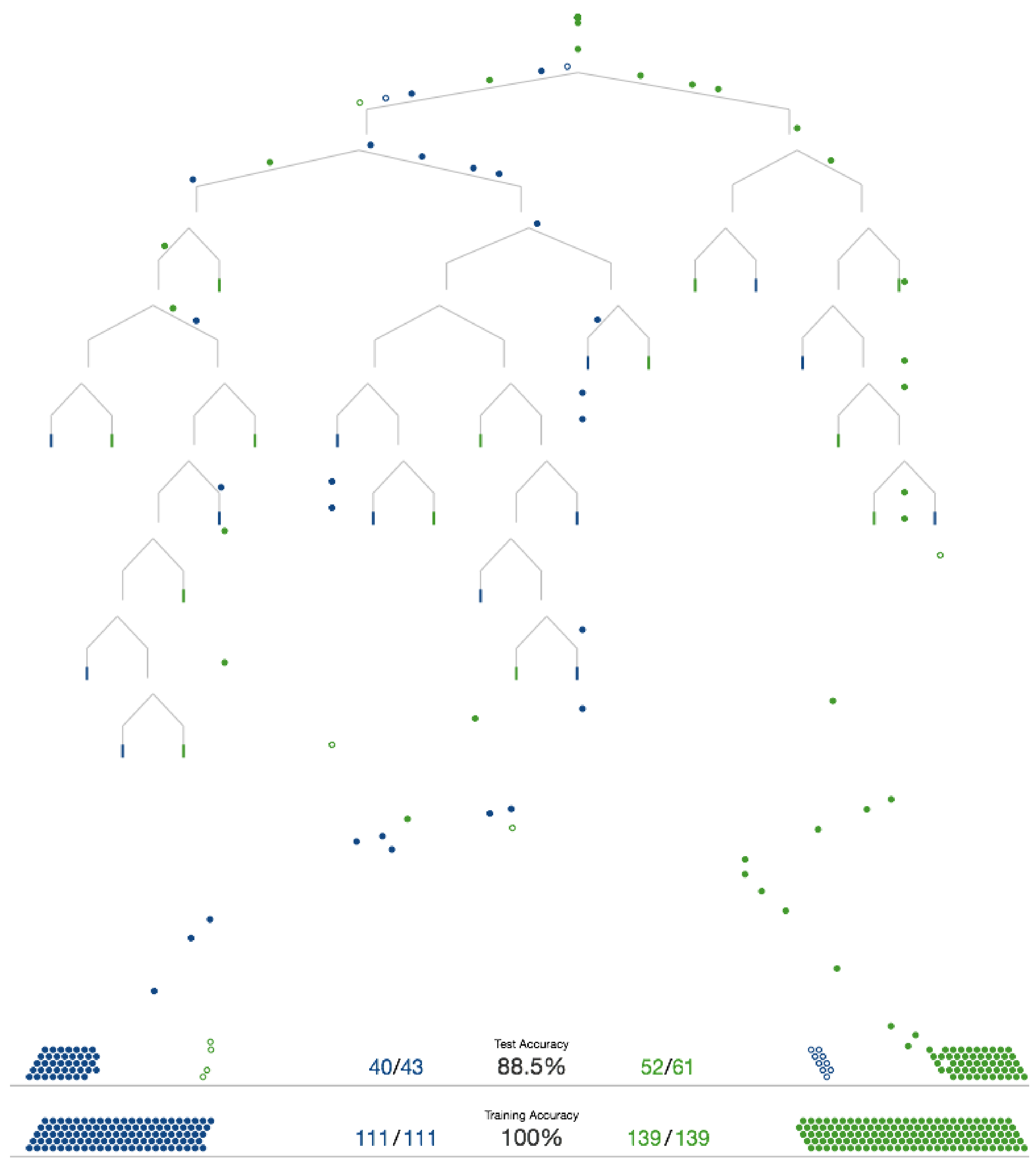
These slides are adopted from Mahdi, Polo, Vivek Srikumar, and Chao Zhang.



X_2
X_1

Visual Introduction to Decision Tree



[Building a tree to distinguish homes in New York from homes in San Francisco]()

Decision Tree: Example (2)

	O	T	H	W	Play?
1	S	H	H	W	-
2	S	H	H	S	-
3	O	H	H	W	+
4	R	M	H	W	+
5	R	C	N	W	+
6	R	C	N	S	-
7	O	C	N	S	+
8	S	M	H	W	-
9	S	C	N	W	+
10	R	M	N	W	+
11	S	M	N	S	+
12	O	M	H	S	+
13	O	H	N	W	+
14	R	M	H	S	-

Outlook: Sunny, Overcast, Rainy

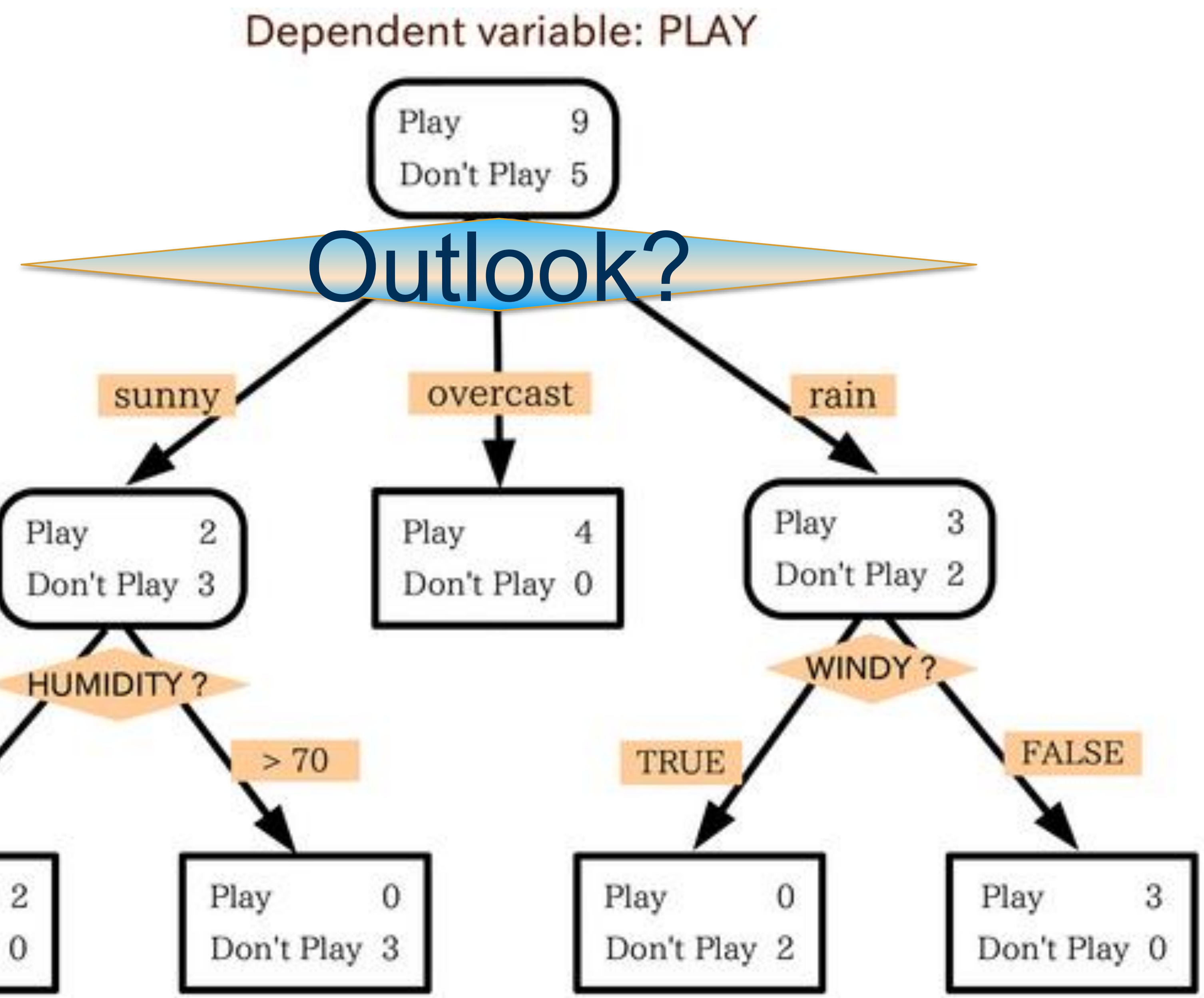
Temperature: Hot, Medium, Cool

Humidity: High, Normal, Low

Wind: Strong, Weak

Will I play tennis today?

Decision trees (DT)



The classifier:

$f_T(x)$: majority class in the leaf in the tree T containing x

Model parameters: The tree structure and size

Decision trees

Pieces:

1. Find the best attribute to split on
2. Find the best split on the chosen attribute
3. Decide on when to stop splitting

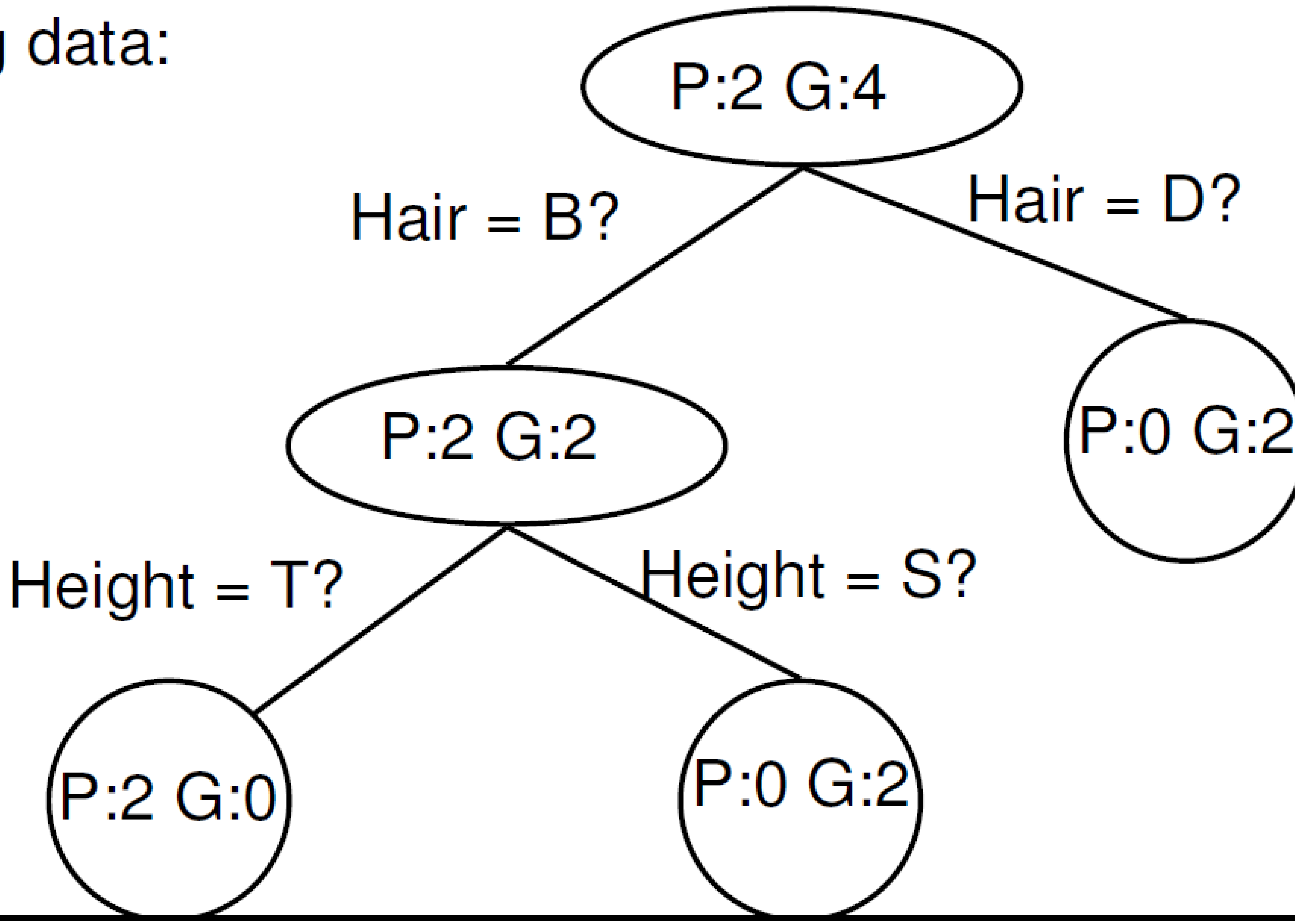
Categorical or Discrete attributes

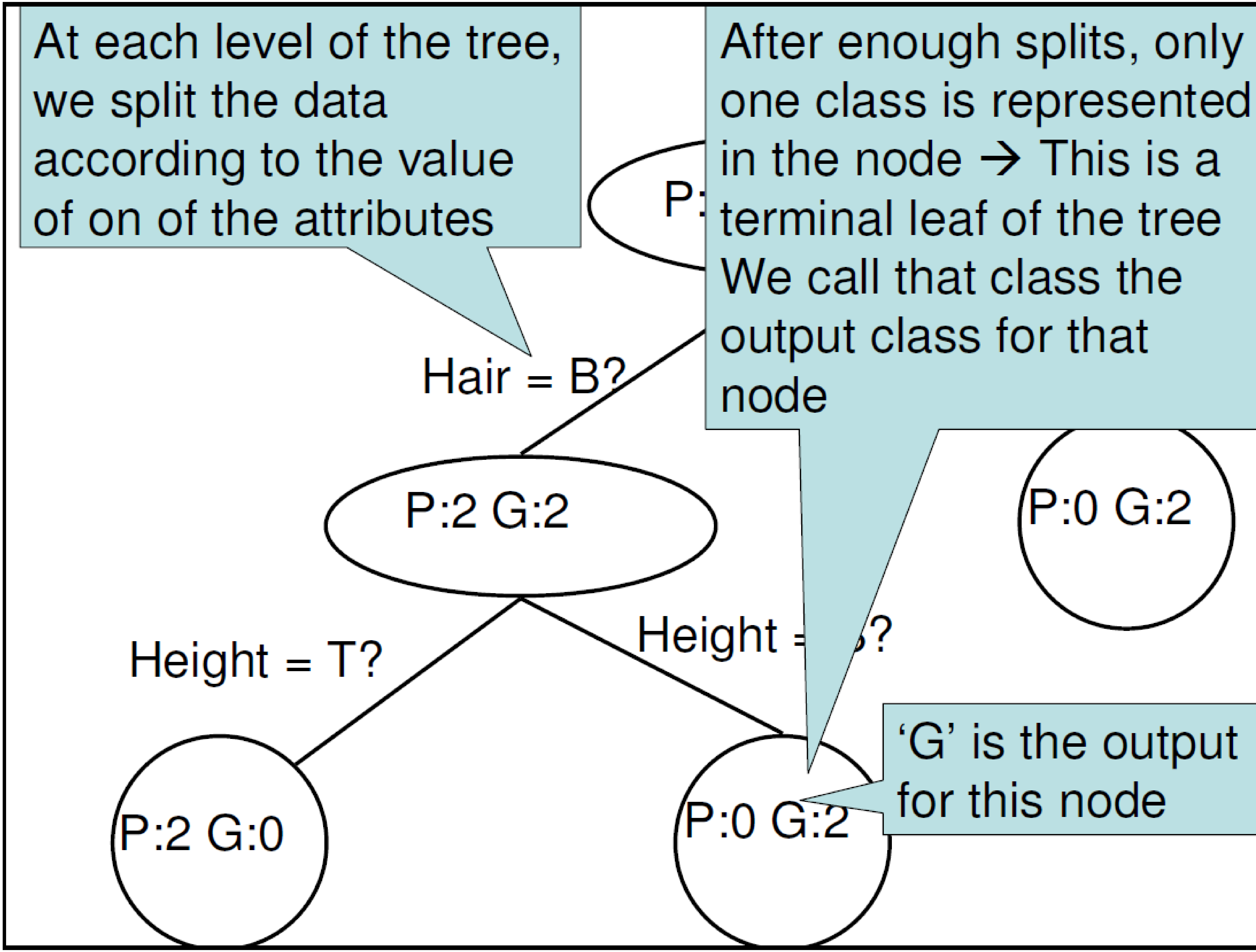
- Three variables:
  - Hair = {blond, dark}
  - Height = {tall,short}
  - Country = {Gromland, Polvia}

Label

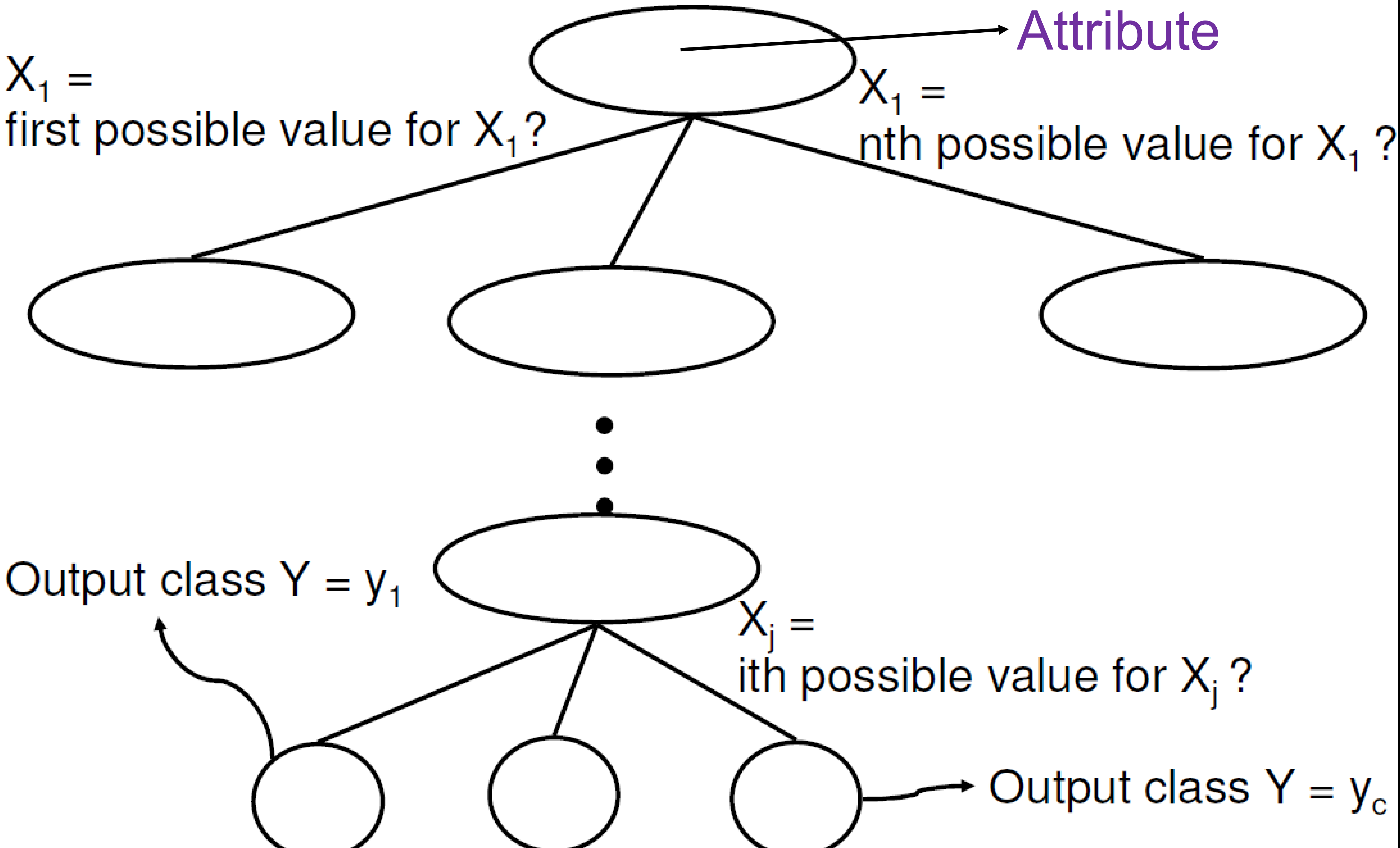
Training data:

(B,T,P)
(B,T,P)
(B,S,G)
(D,S,G)
(D,T,G)
(B,S,G)

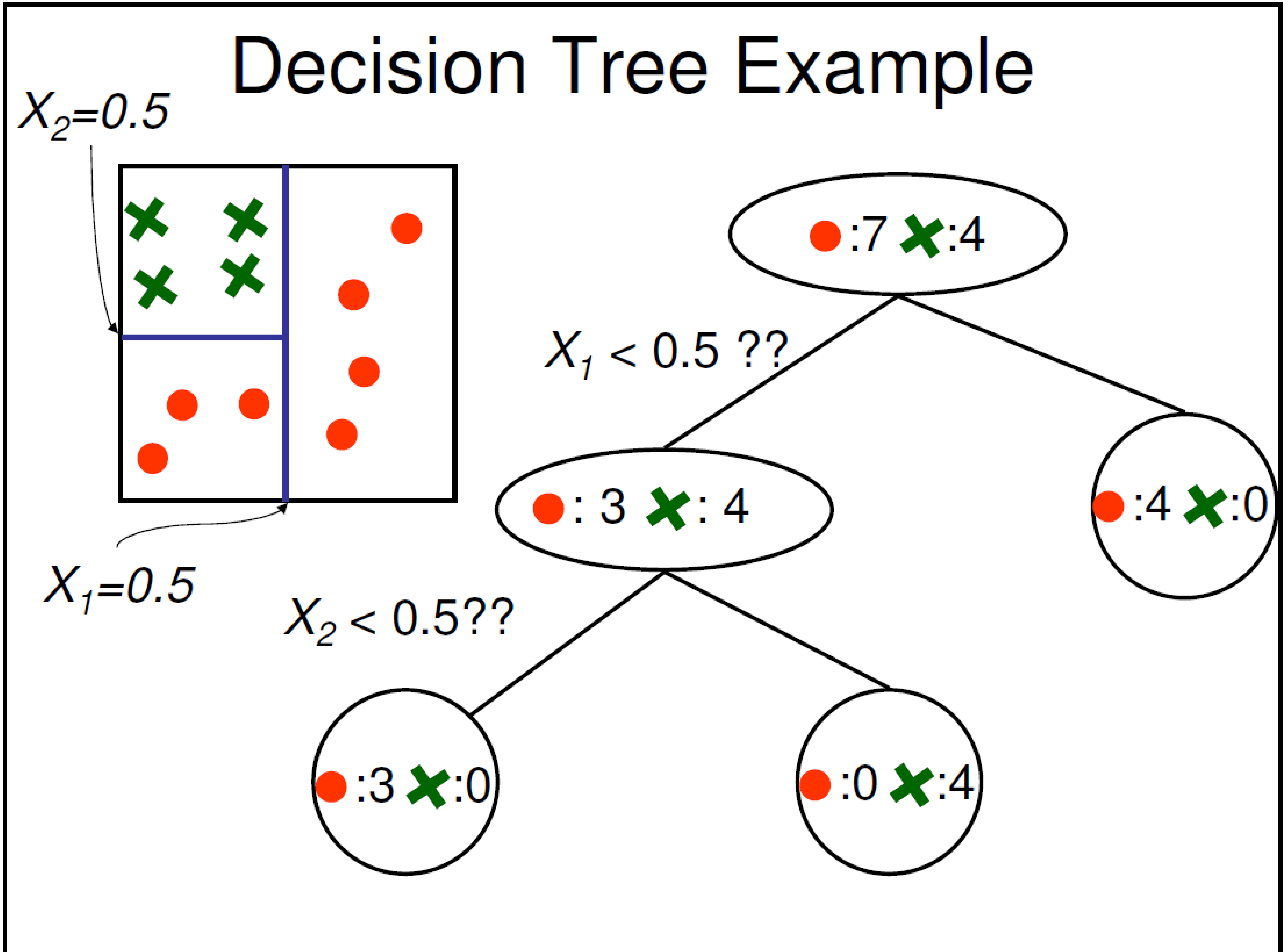



At each level of the tree, we split the data according to the value of on of the attributes
After enough splits, only one class is represented in the node → This is a terminal leaf of the tree We call that class the output class for that node
P:
Hair = B?
P:2 G:2
P:0 G:2
Height = T?
Height =
?
'G' is the output for this node
P:2 G:0
P:0 G:2

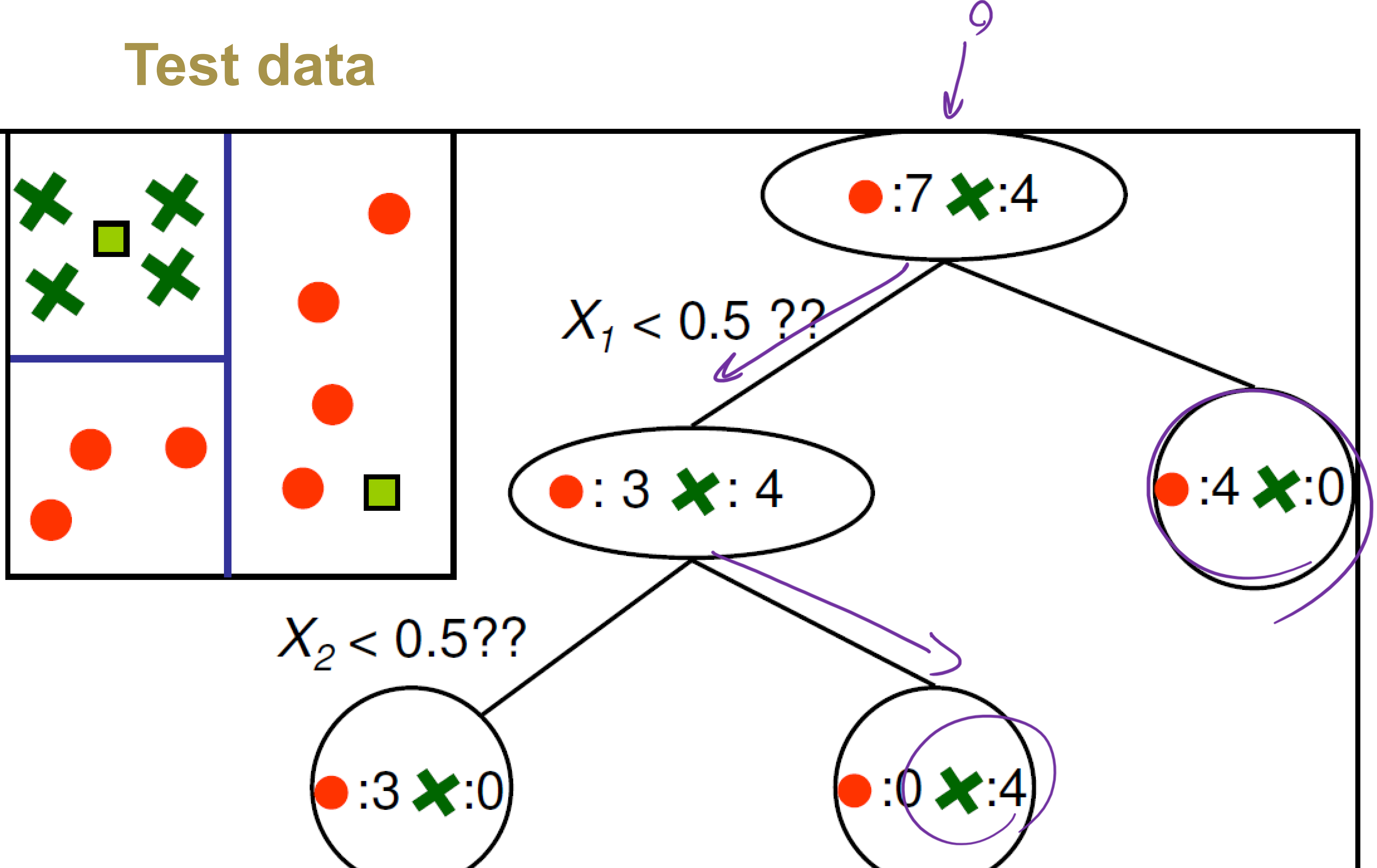
General Decision Tree (Discrete Attributes)



Continuous attributes



Test data

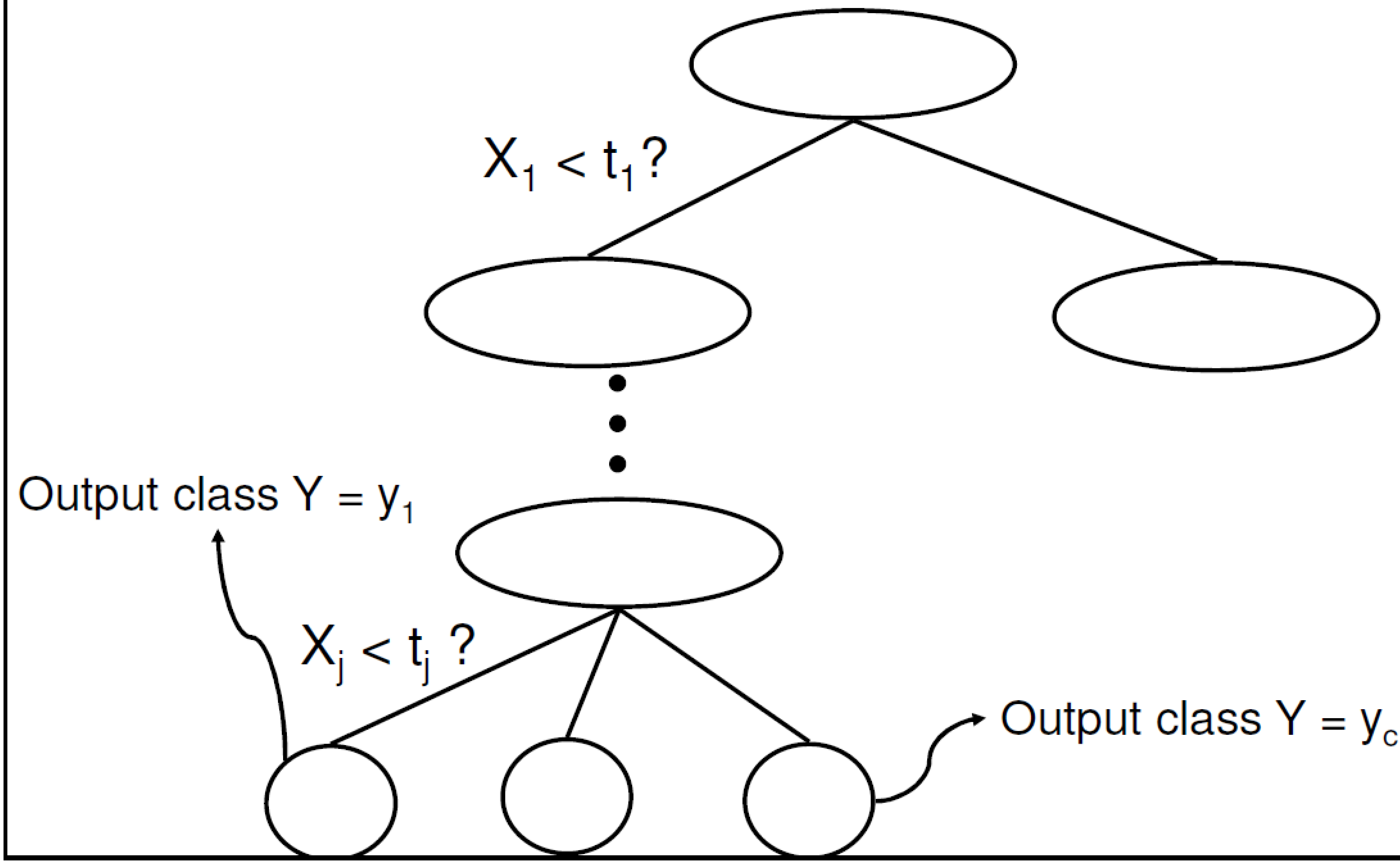


The class of a new input can be classified by following the tree all the way down to a leaf and by reporting the output of the leaf. For example:

(0.2,0.8) is classified as

(0.8,0.2) is classified as

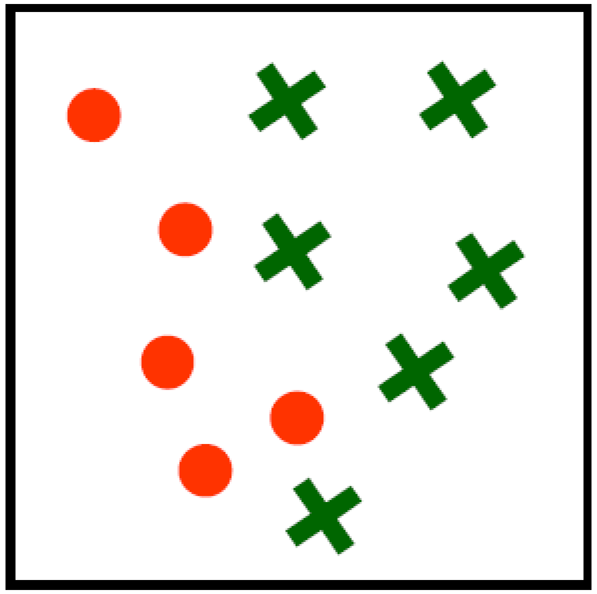
General Decision Tree (Continuous Attributes)



Basic Questions

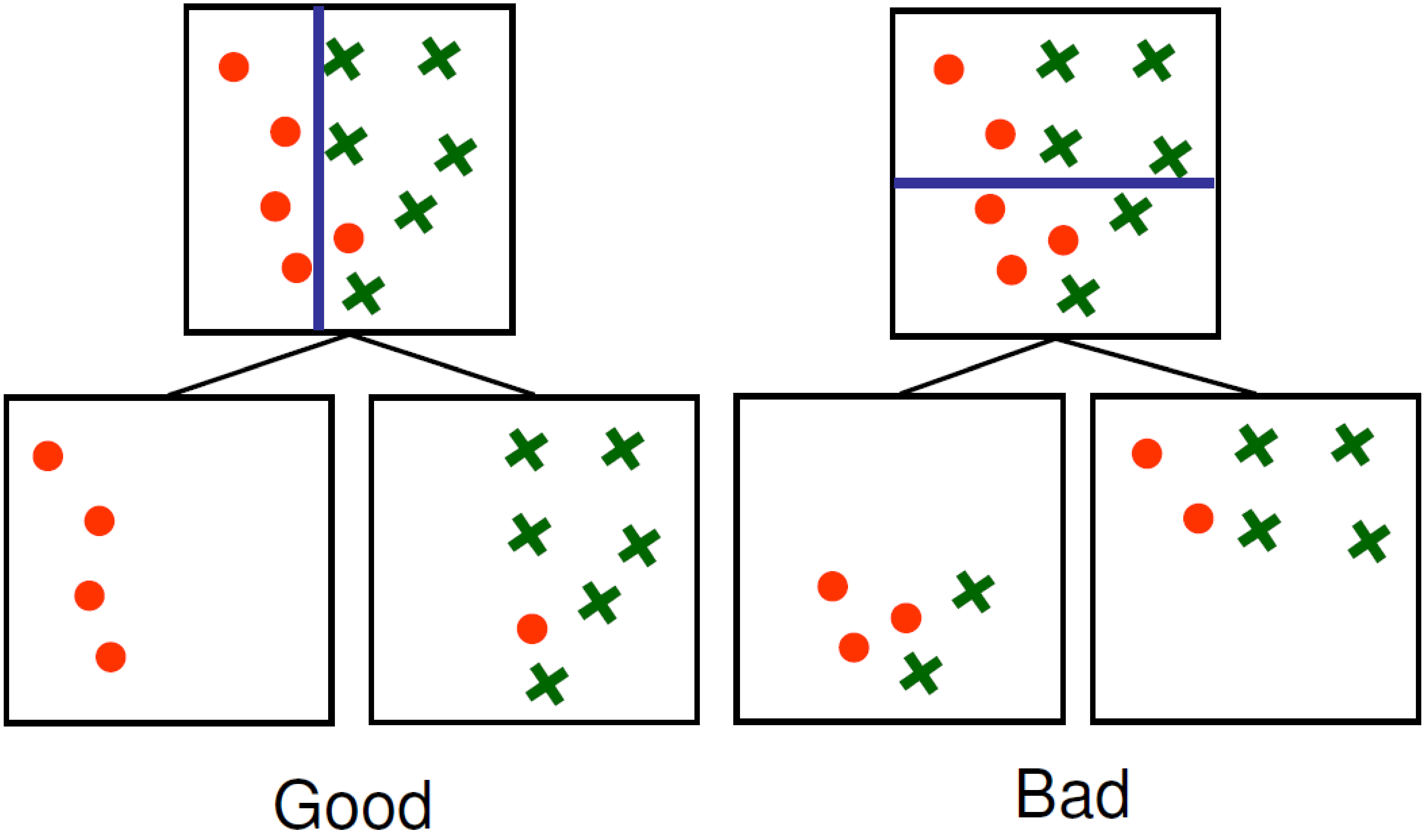
- How to choose the attribute/value to split on at each level of the tree?
- When to stop splitting? When should a node be declared a leaf?
- If a leaf node is impure, how should the class label be assigned?
- If the tree is too large, how can it be pruned?

How to choose the attribute/value to split on at each level of the tree?

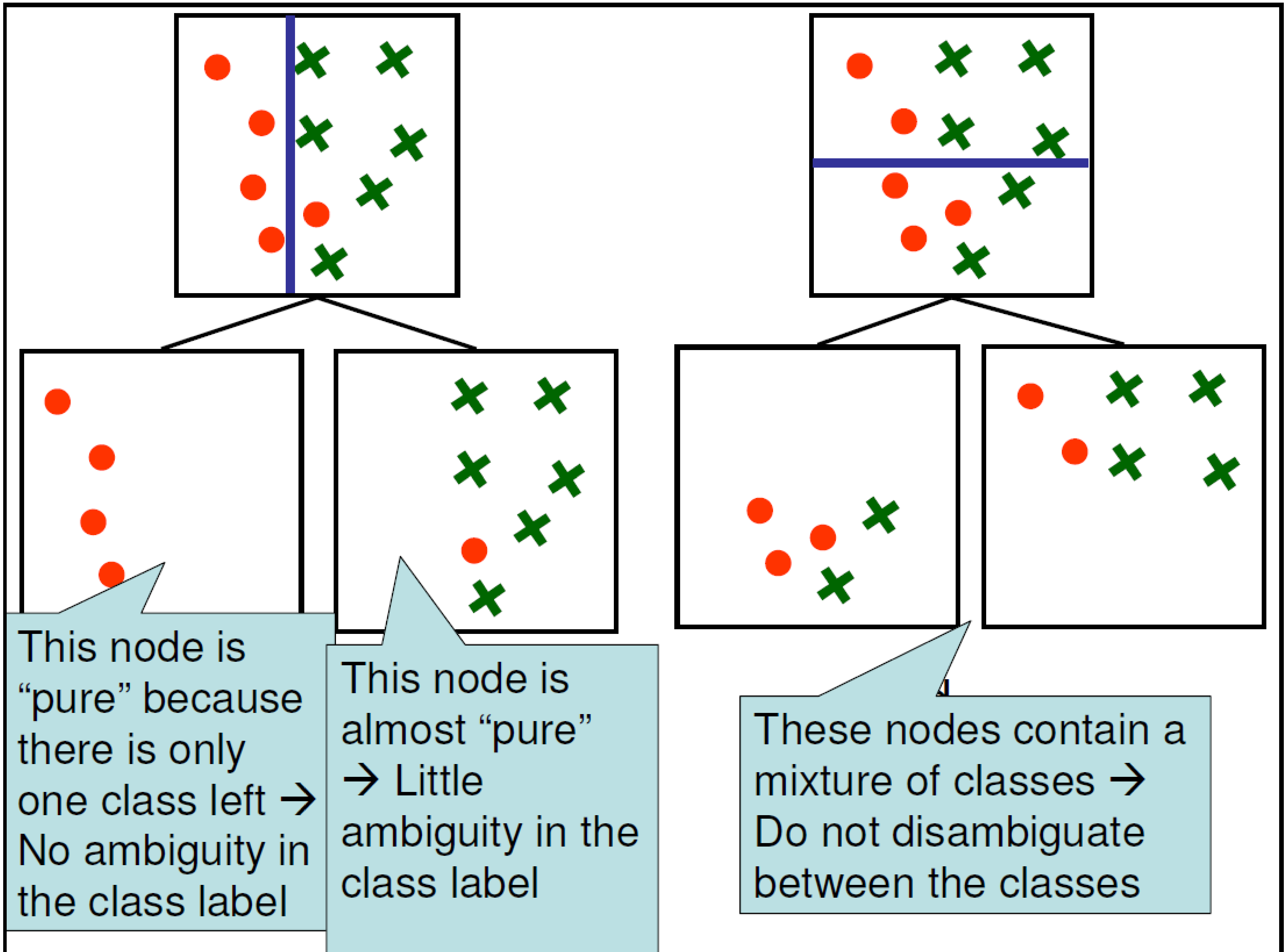


- Two classes (red circles/green crosses)
- Two attributes: X_1 and X_2
- 11 points in training data
- Idea → Construct a decision tree such that the leaf nodes predict correctly the class for all the training examples

How to choose the attribute/value to split on at each level of the tree?



We want to find the most compact, smallest size tree (Occam's razor), that classifies the training data correctly → We want to find the split choices that will get us the fastest to pure nodes



Information Theory Recap

- I(x)

- H(x)

- Binary Case: Maximum value of entropy is

- Non-binary case: Maximum value of entropy is

- Range of entropy

Information Content

Coin flip

C_{1H}	0
C_{1T}	6

P(C_{1H}) = 0/6 = 0

P(C_{1T}) = 6/6 = 1

C_{2H}	1
C_{2T}	5

P(C_{2H}) = 1/6

P(C_{2T}) = 5/6

C_{3H}	2
C_{3T}	4

P(C_{3H}) = 2/6

P(C_{3T}) = 4/6

Which coin will give us the purest information?

Entropy ~ Uncertainty

Lower uncertainty, higher information gain

$$\boxed{H(X) = -\sum_{i=1}^{N} P(x=i)\log_2 P(x=i)}$$

Entropy = – 0 log 0 – 1 log 1 = – 0 – 0 = 0

Entropy = – (1/6) $\log_2$ (1/6) – (5/6) $\log_2$ (5/6) = 0.65

Entropy = – (2/6) $\log_2$ (2/6) – (4/6) $\log_2$ (4/6) = 0.92

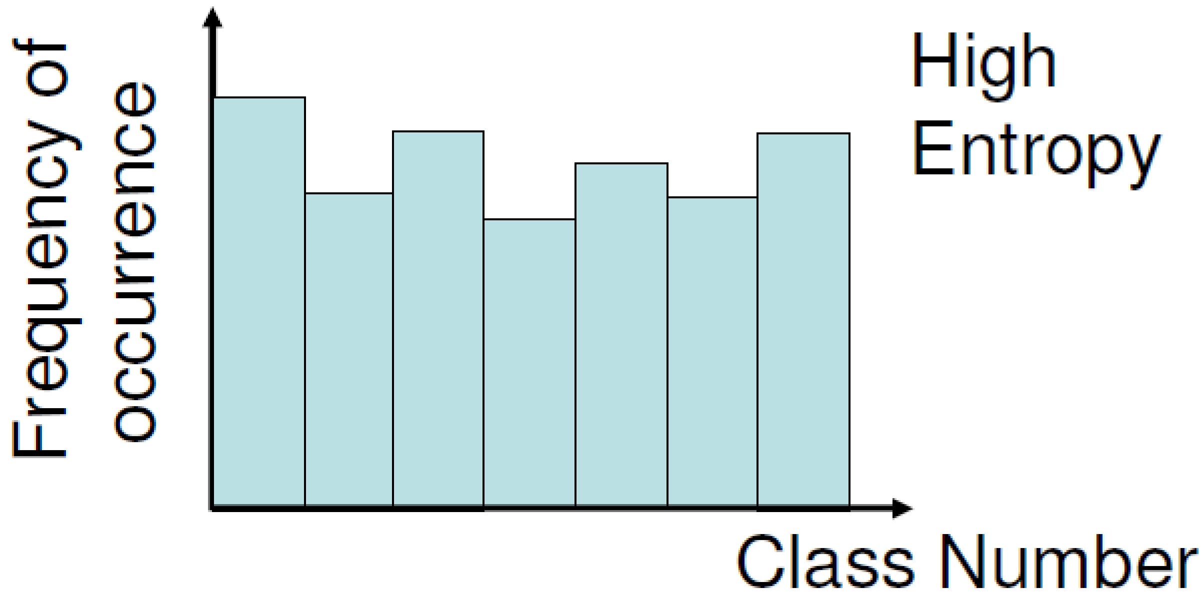
Entropy

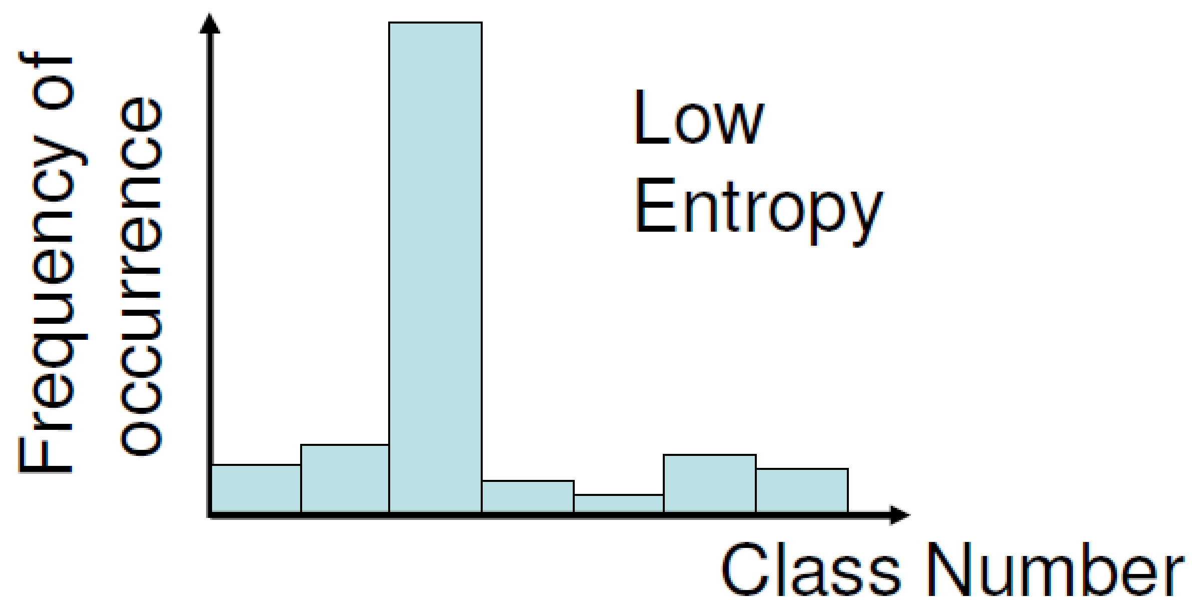
- In general, the average number of bits necessary to encode *n* values is the entropy:

$$\boxed{H = -\sum_{i=1}^{n} P_i \log_2 P_i}$$

- P_i = probability of occurrence of value i
  - High entropy → All the classes are (nearly) equally likely
  - Low entropy → A few classes are likely; most of the classes are rarely observed

Entropy

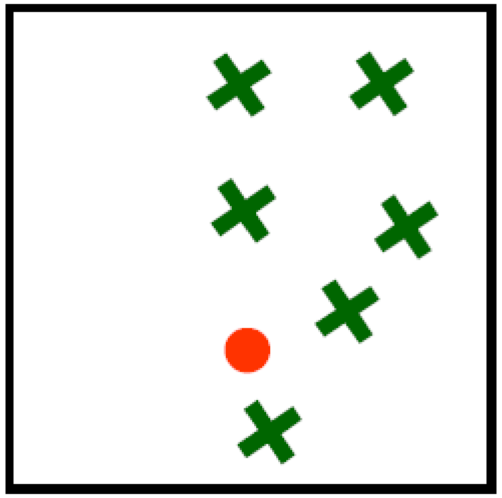




The entropy captures the degree of “purity” of the distribution

Example Entropy Calculation

1

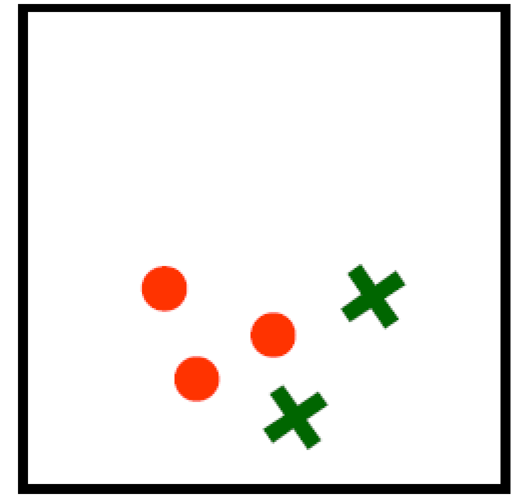


$N_A = 1$
$N_B = 6$

$p_A = N_A/(N_A+N_B) = 1/7$
$p_B = N_B/(N_A+N_B) = 6/7$

$H_1 = -p_A \log_2 p_A - p_B \log_2 p_B$
$= 0.59$

2



$N_A = 3$
$N_B = 2$

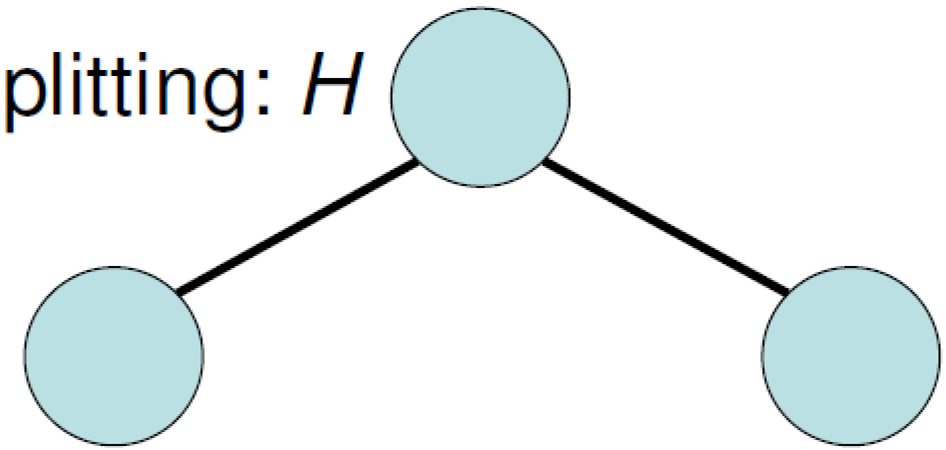
$p_A = N_A/(N_A+N_B) = 3/5$
$p_B = N_B/(N_A+N_B) = 2/5$

$H_2 = -p_A \log_2 p_A - p_B \log_2 p_B$
$= 0.97$

$H_1 < H_2$ => (2) less pure than (1)

Conditional Entropy

Entropy before splitting: H



After splitting, a fraction P_L of the data goes to the left node, which has entropy H_L

After splitting, a fraction P_R of the data goes to the left node, which has entropy H_R

The average entropy after splitting is:

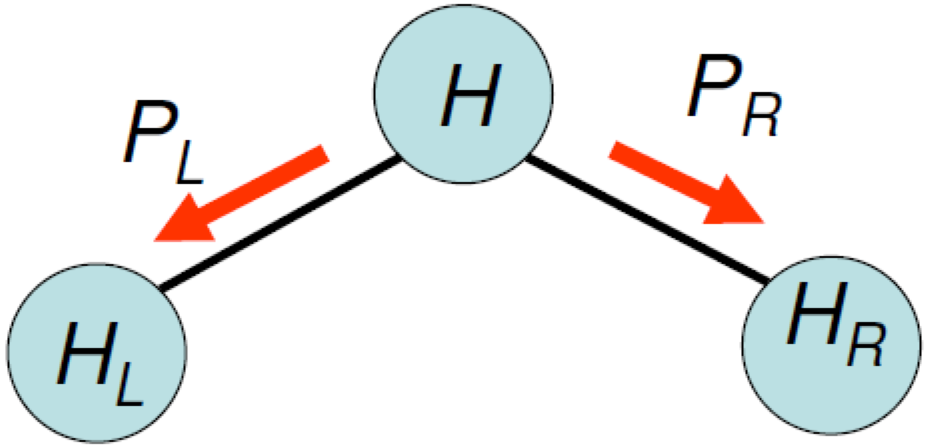
Entropy of left node

$$\boxed{H_L \times P_L + H_R \times P_R}$$

"Conditional Entropy"

Probability that a random input is directed to the left node

Information Gain



We want nodes as pure as possible

→We want to reduce the entropy as much as possible

→ We want to maximize the difference between the entropy of the parent node and the expected entropy of the children

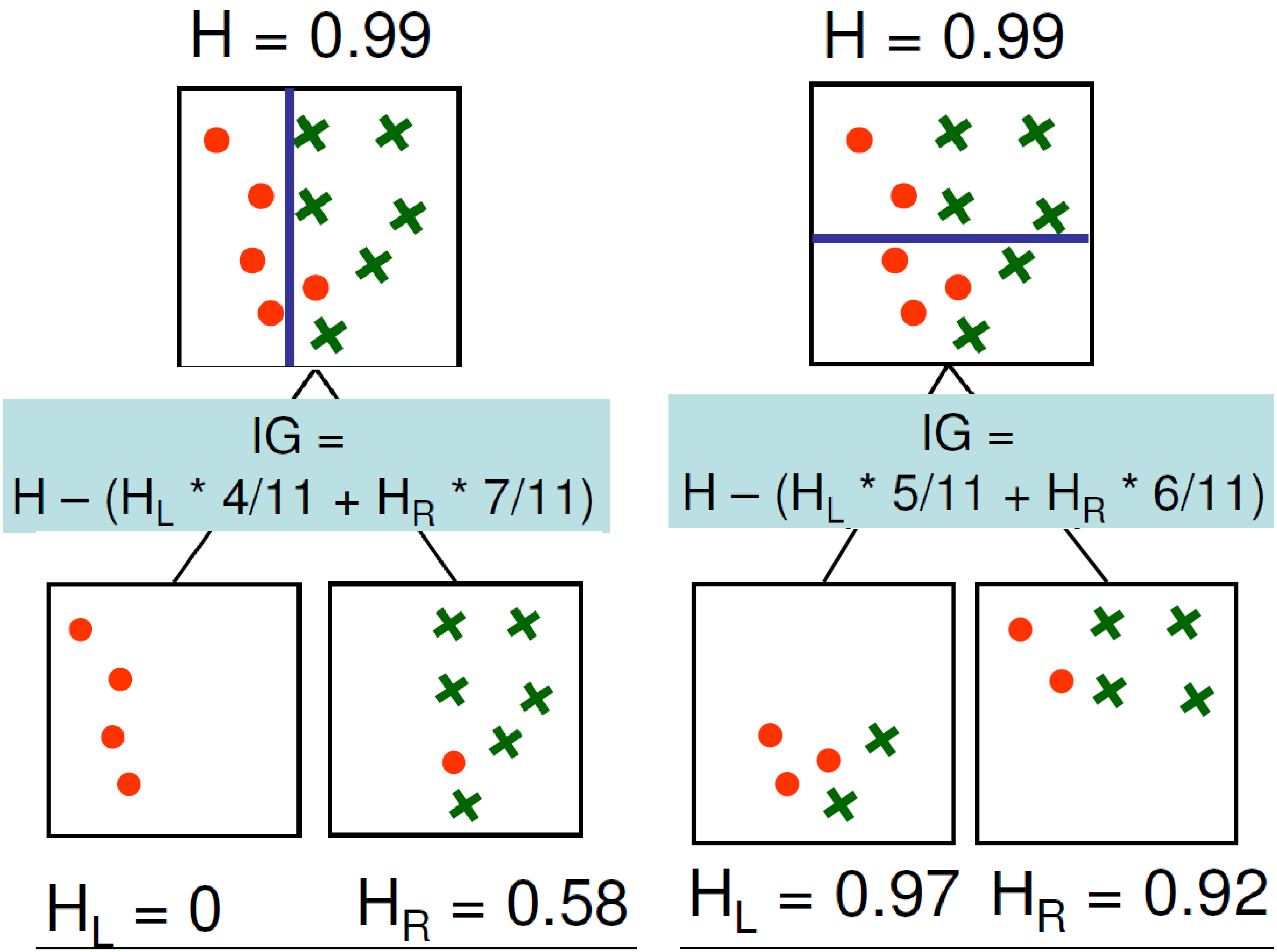
Information Gain (IG) = Amount by which the ambiguity is decreased by splitting the node

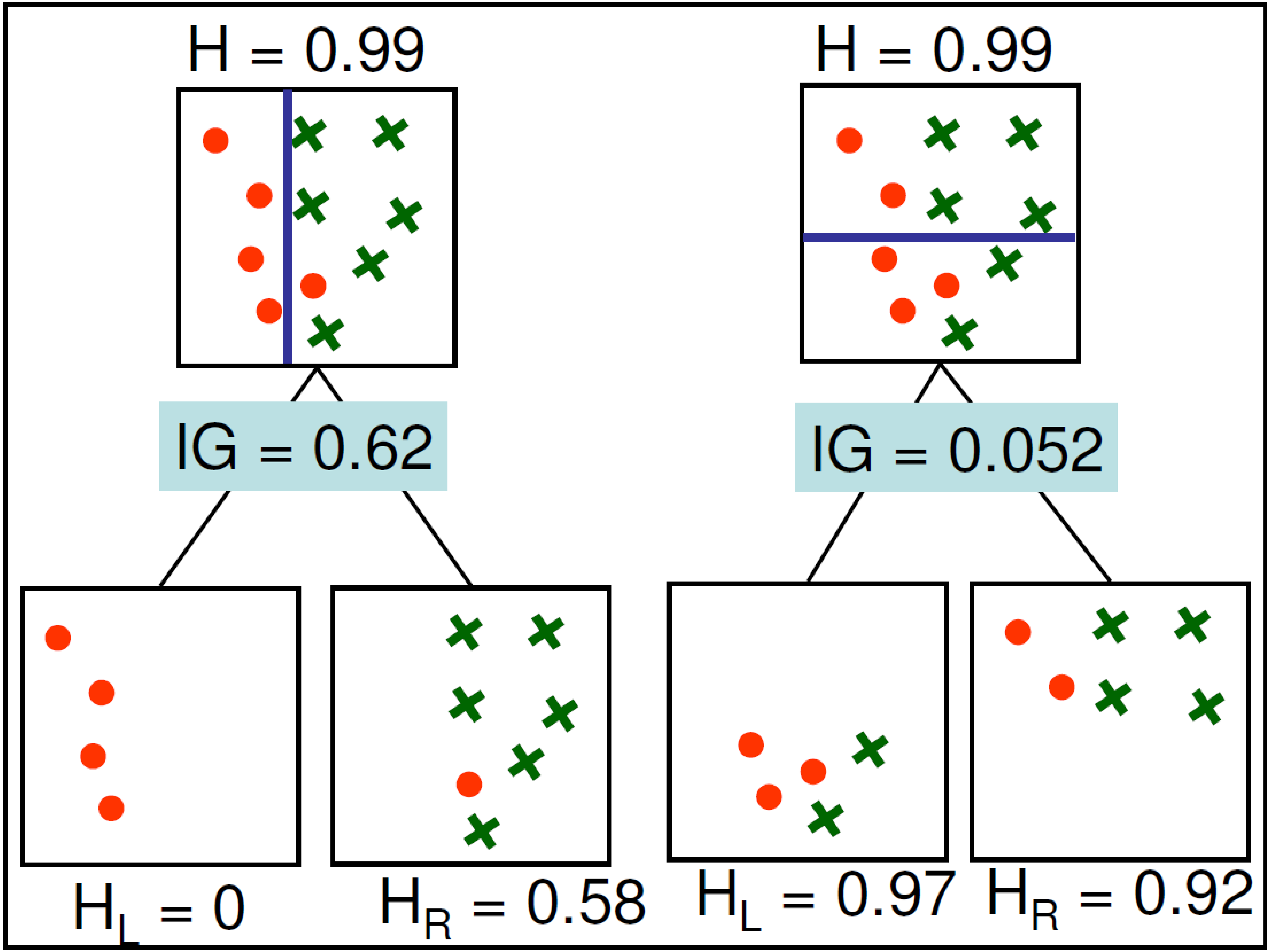
Maximize:

$$\boxed{IG = H - (H_L \times P_L + H_R \times P_R)}$$

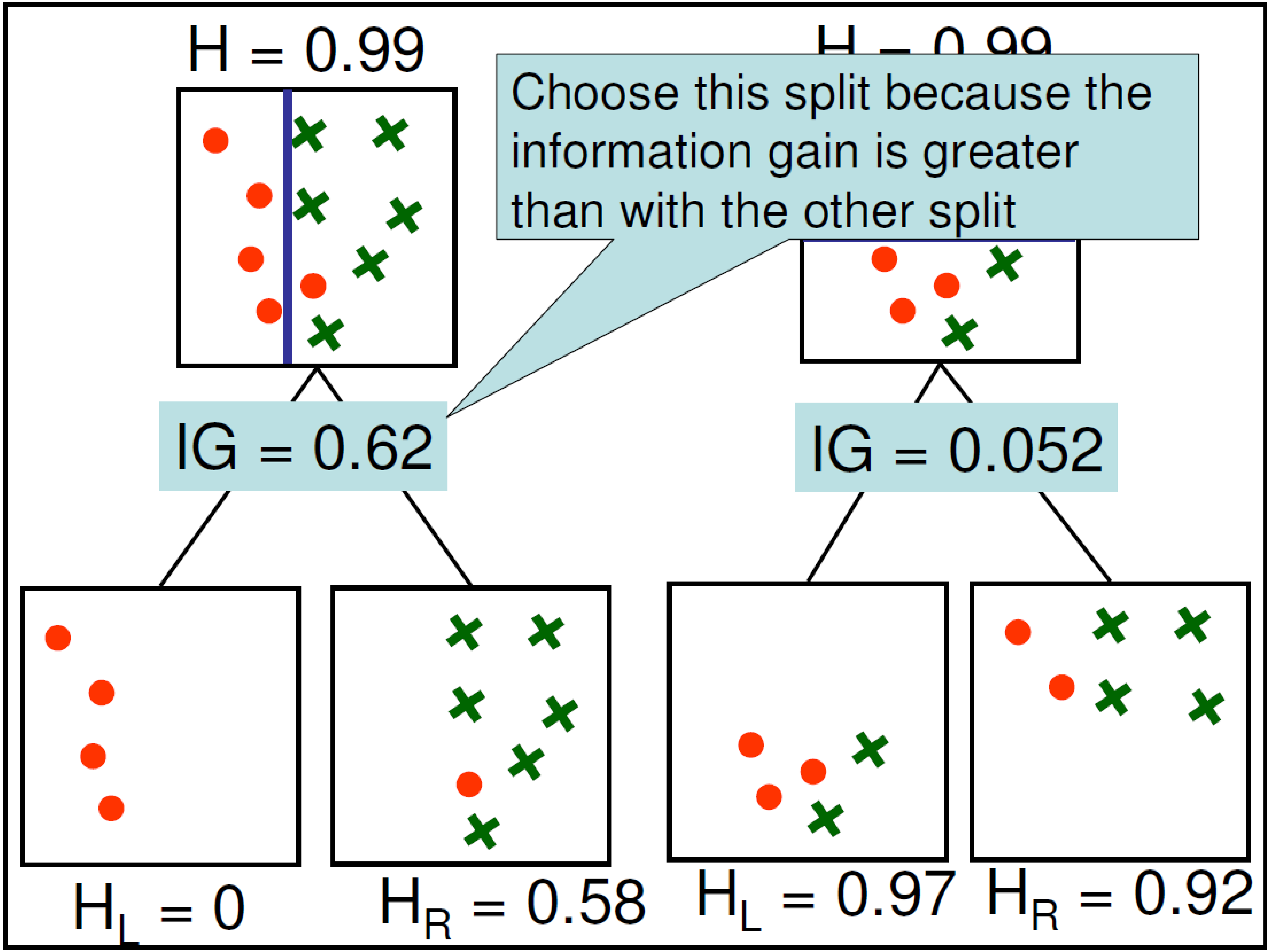
Notations

- Entropy: $H(Y)$ = Entropy of the distribution of classes at a node
- Conditional Entropy:
  - *Discrete*: $H(Y|X_j)$ = Entropy after splitting with respect to variable j
  - *Continuous*: $H(Y|X_j,t)$ = Entropy after splitting with respect to variable j with threshold t
- Information gain:
  - *Discrete*: $IG(Y|X_j) = H(Y) - H(Y|X_j)$ = Entropy after splitting with respect to variable j
  - *Continuous*: $IG(Y|X_j,t) = H(Y) - H(Y|X_j,t)$ = Entropy after splitting with respect to variable j with threshold t


H = 0.99
IG =
H – (H_L * 4/11 + H_R * 7/11)
H_L = 0
H_R = 0.58
H = 0.99
IG =
H – (H_L * 5/11 + H_R * 6/11)
H_L = 0.97
H_R = 0.92



H = 0.99
H = 0.99
IG = 0.62
IG = 0.052
H_L = 0
H_R = 0.58
H_L = 0.97
H_R = 0.92



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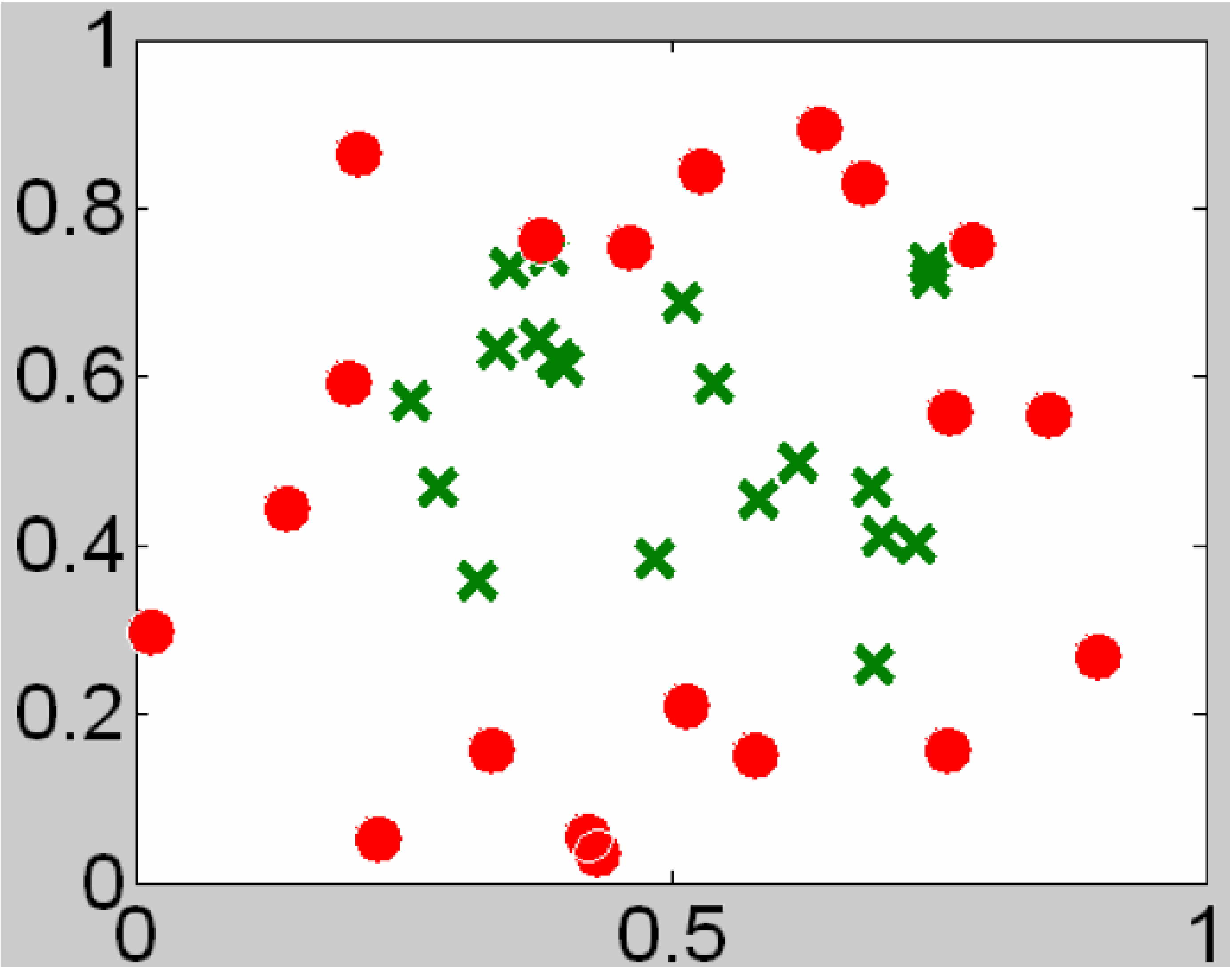
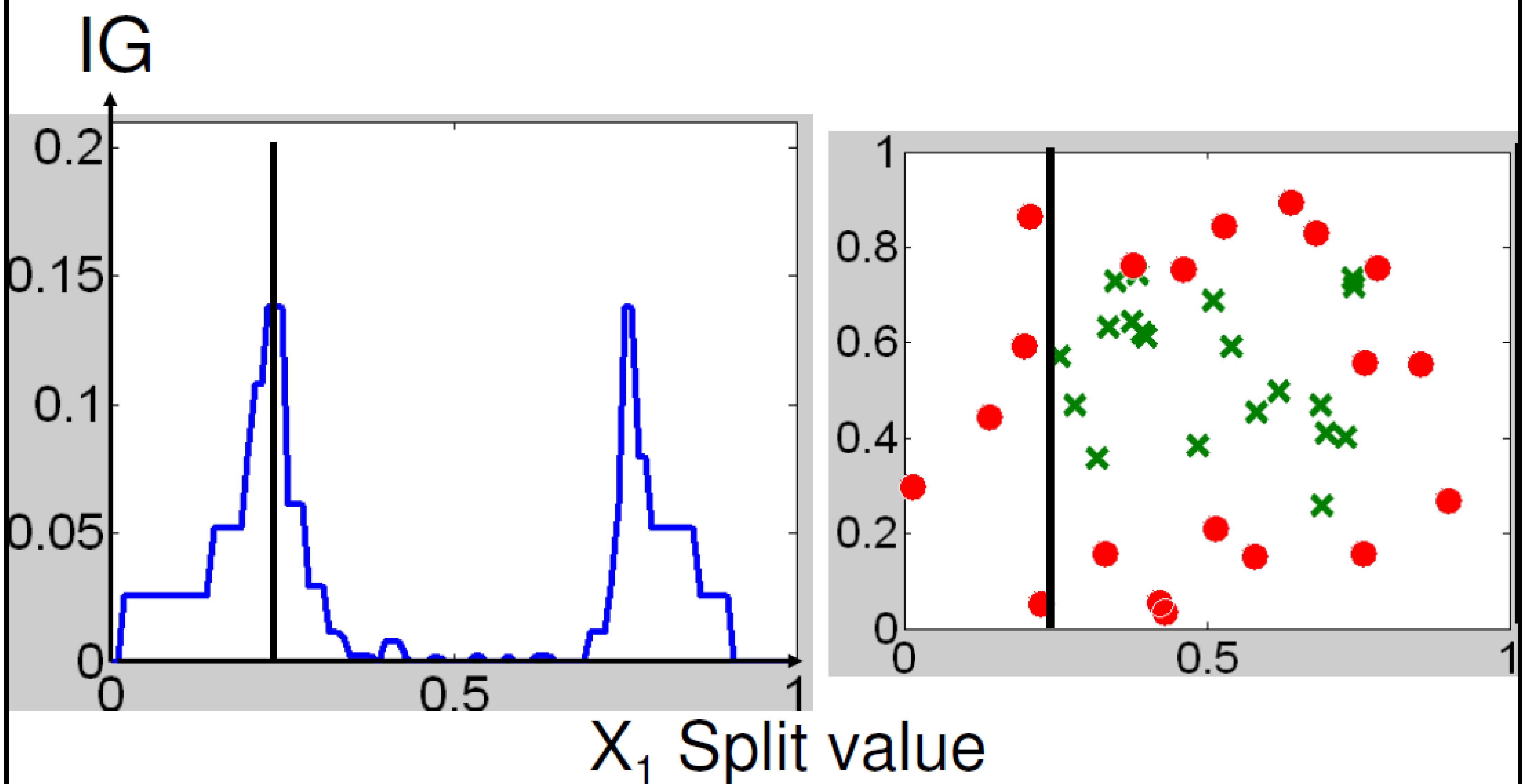
More Complete Example



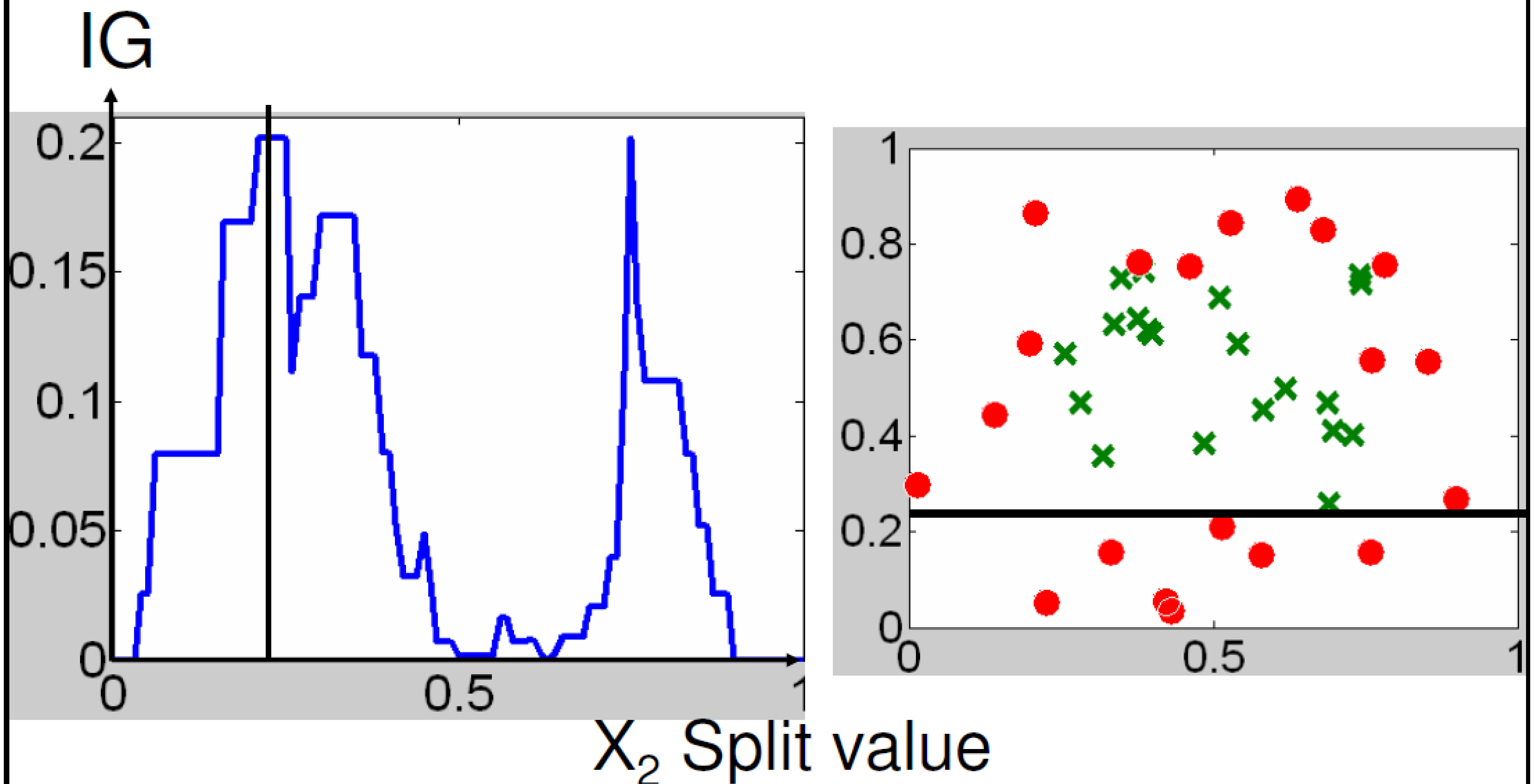
● = 20 training examples from class A

✖ = 20 training examples from class B

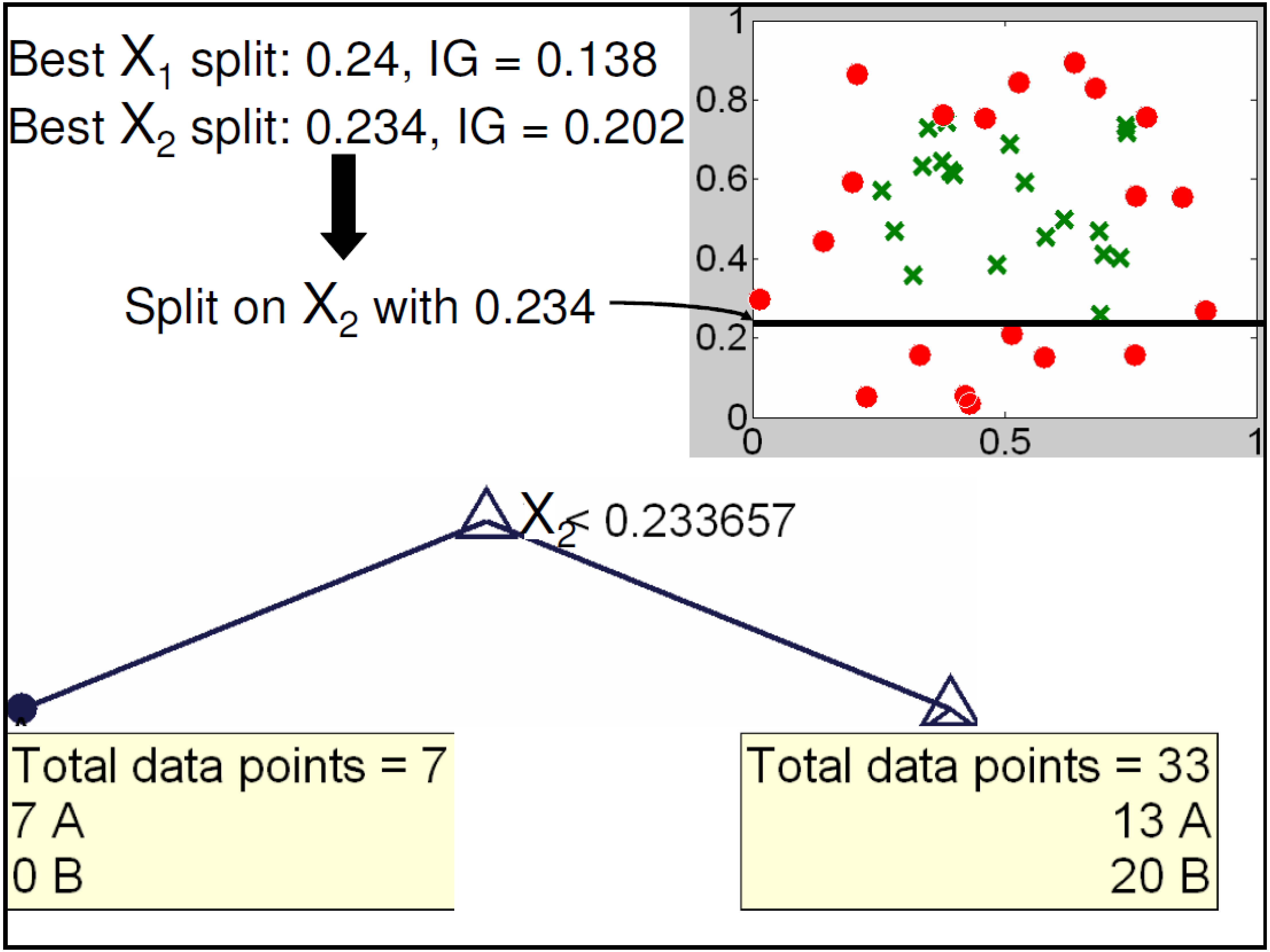
Attributes = X_1 and X_2 coordinates



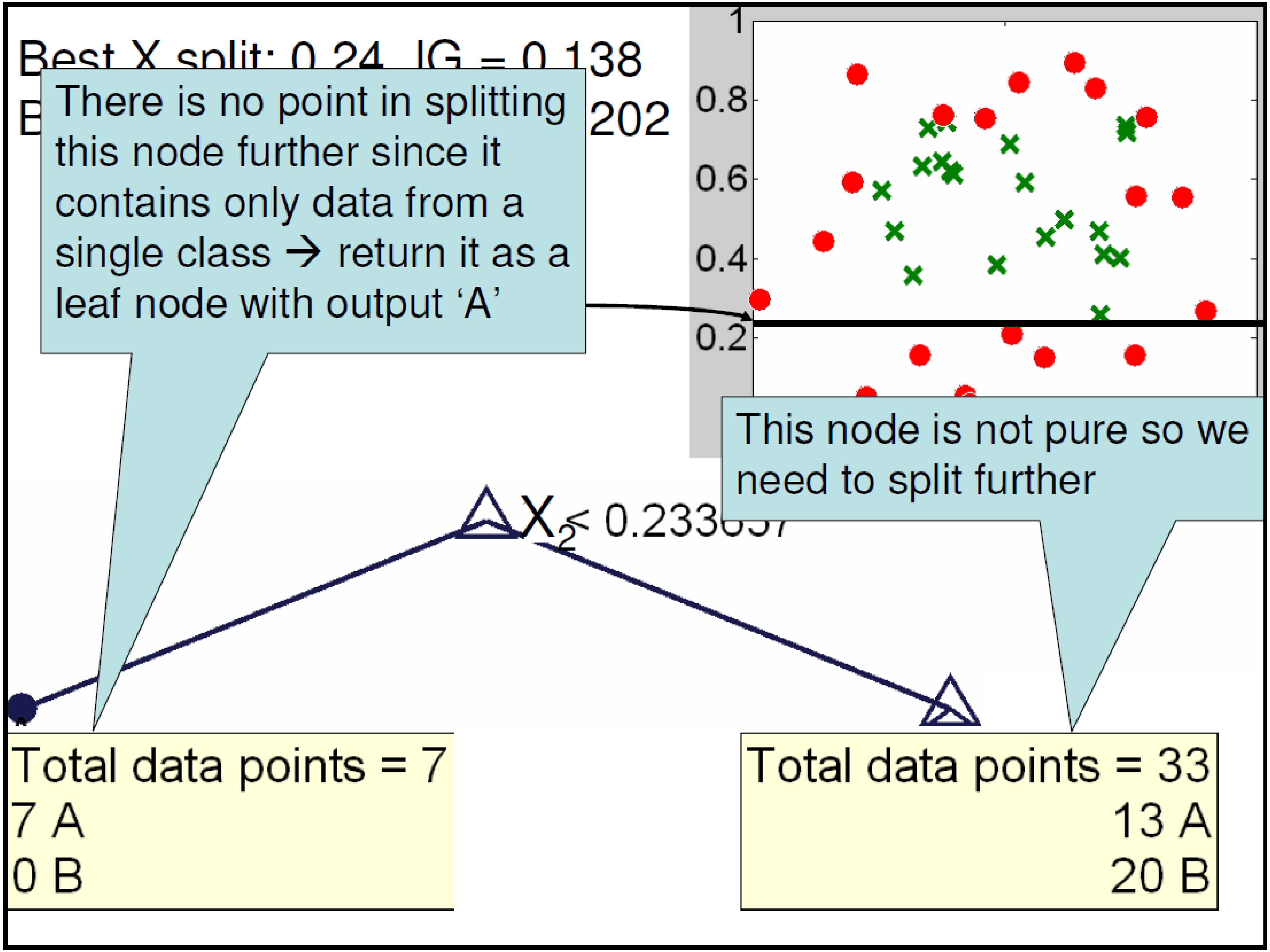
Best split value (max Information Gain) for X_1 attribute: 0.24 with IG = 0.138

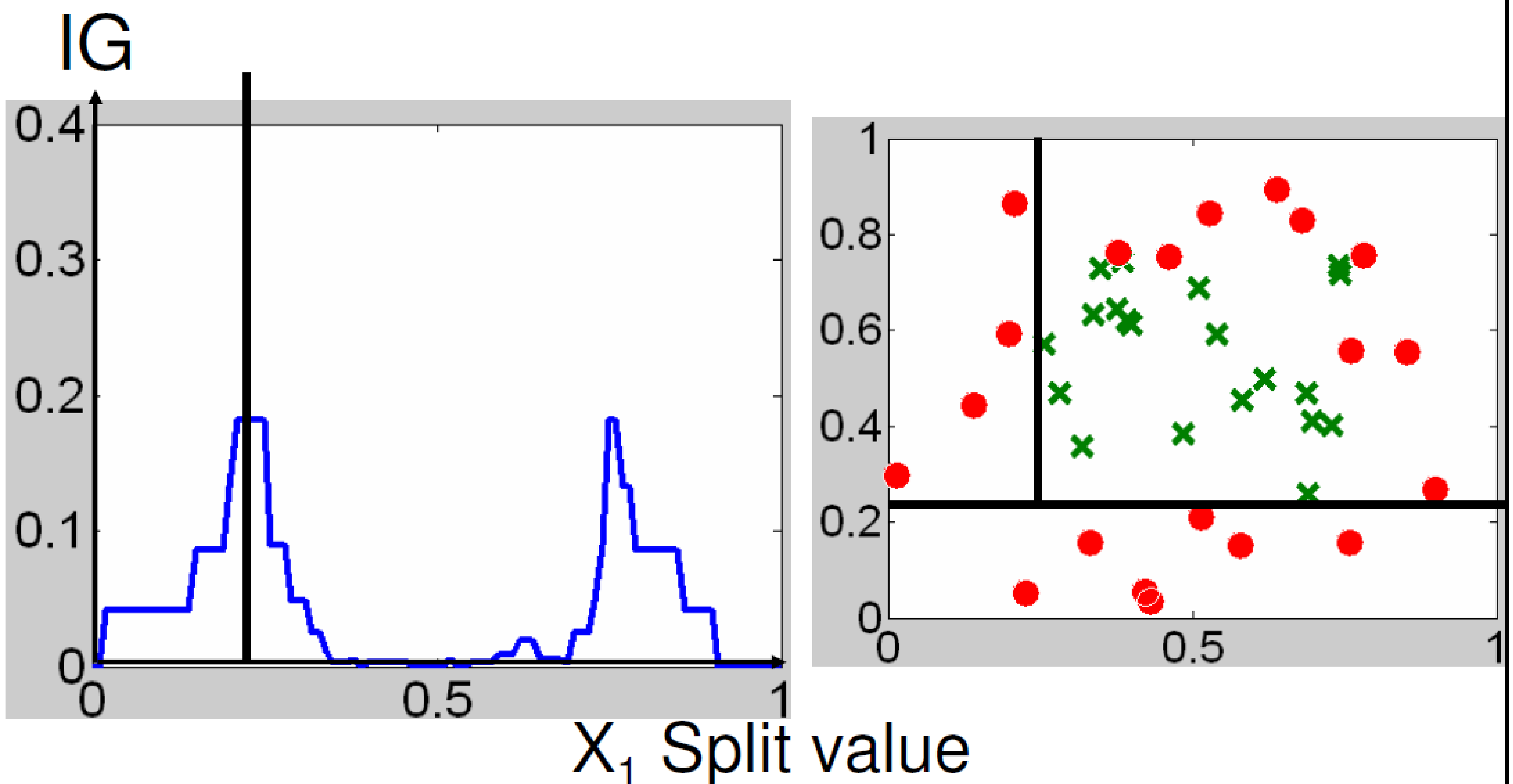


Best split value (max Information Gain) for X_2 attribute: 0.234 with IG = 0.202

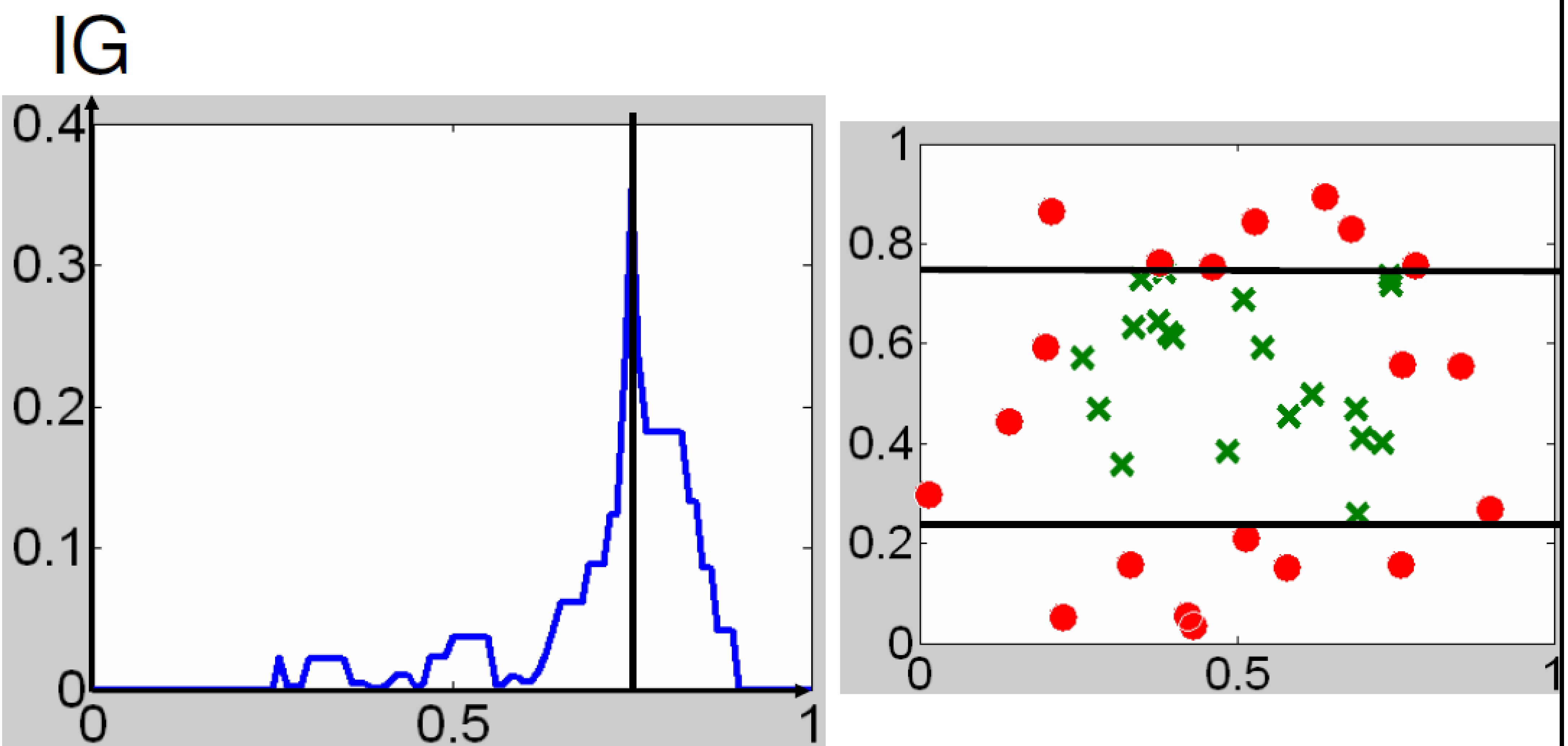


Left direction for **smaller** value, **right** direction for **bigger** value


Best X split: 0.24, IG = 0.138
202
There is no point in splitting this node further since it contains only data from a single class → return it as a leaf node with output 'A'
This node is not pure so we need to split further
1
0.8
0.6
0.4
0.2
X2 < 0.233
Total data points = 7
7 A
0 B
Total data points = 33
13 A
20 B



Best split value (max Information Gain) for X_1 attribute: 0.22 with IG ~ 0.182



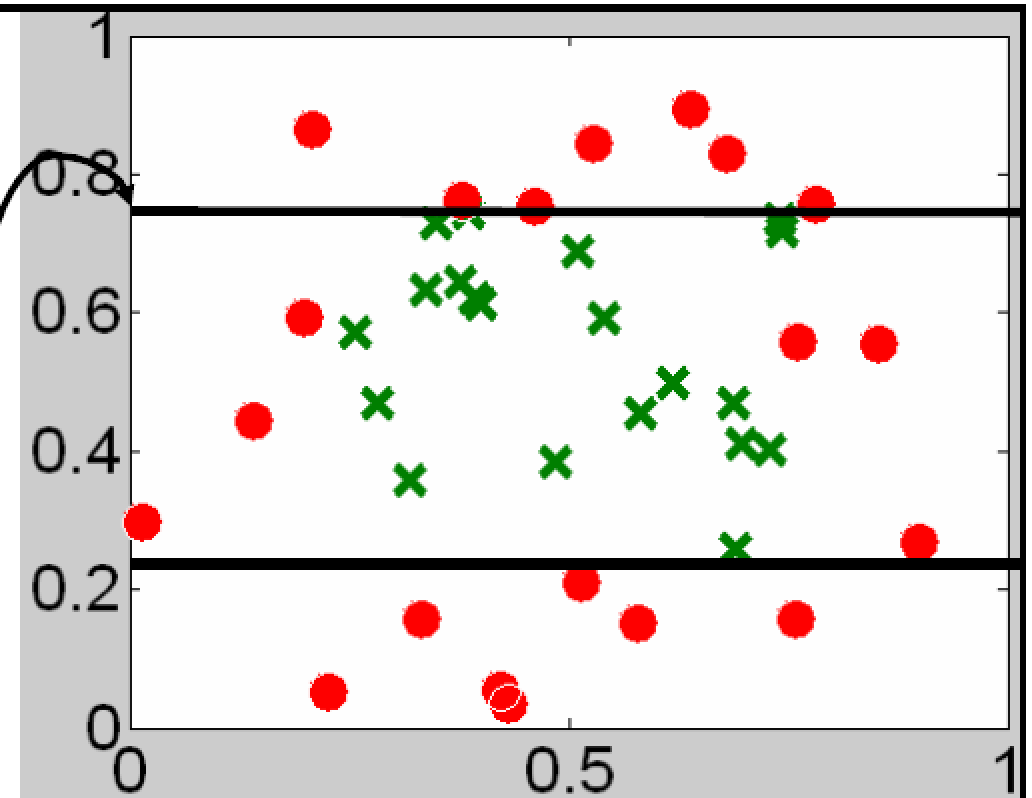
X_2 Split value

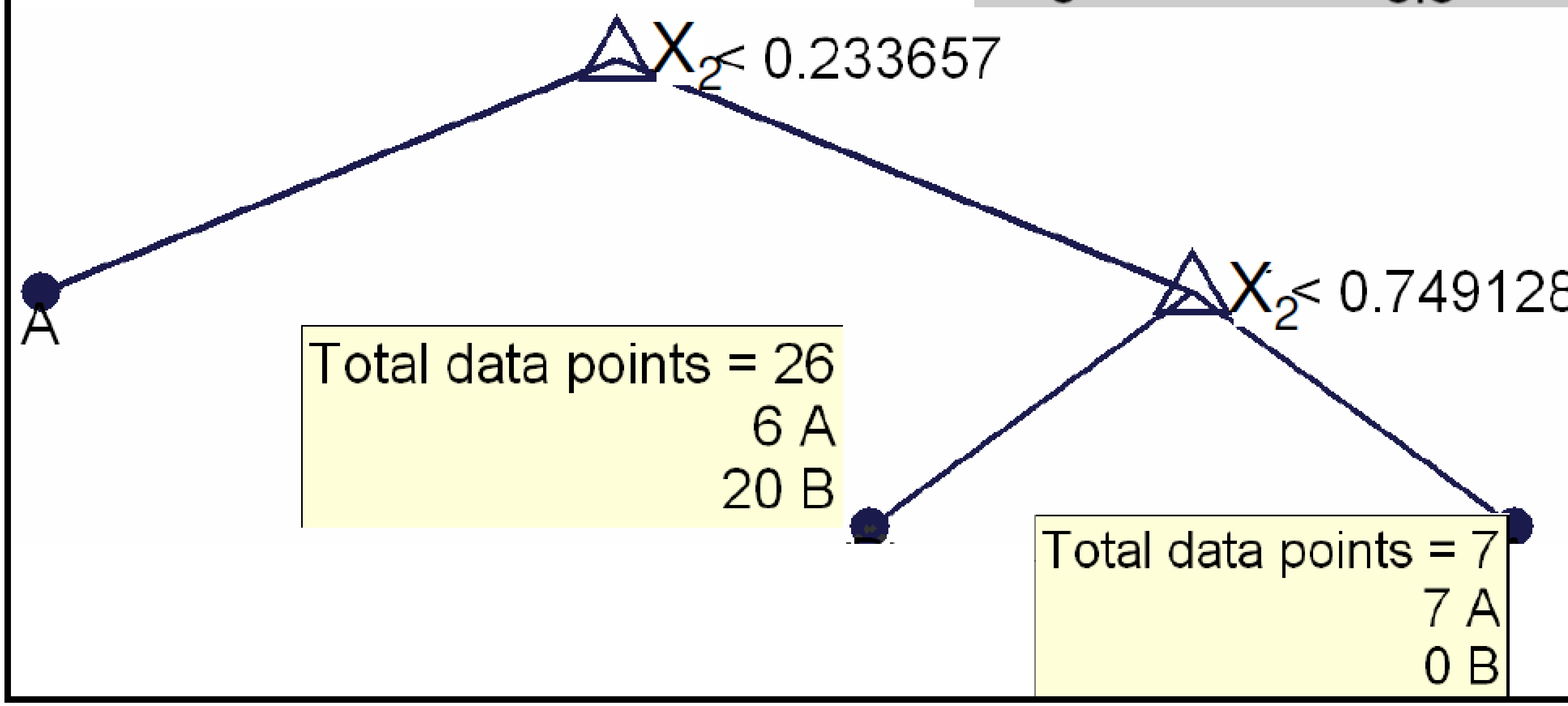
Best split value (max Information Gain) for X_2 attribute: 0.75 with IG ~ 0.353

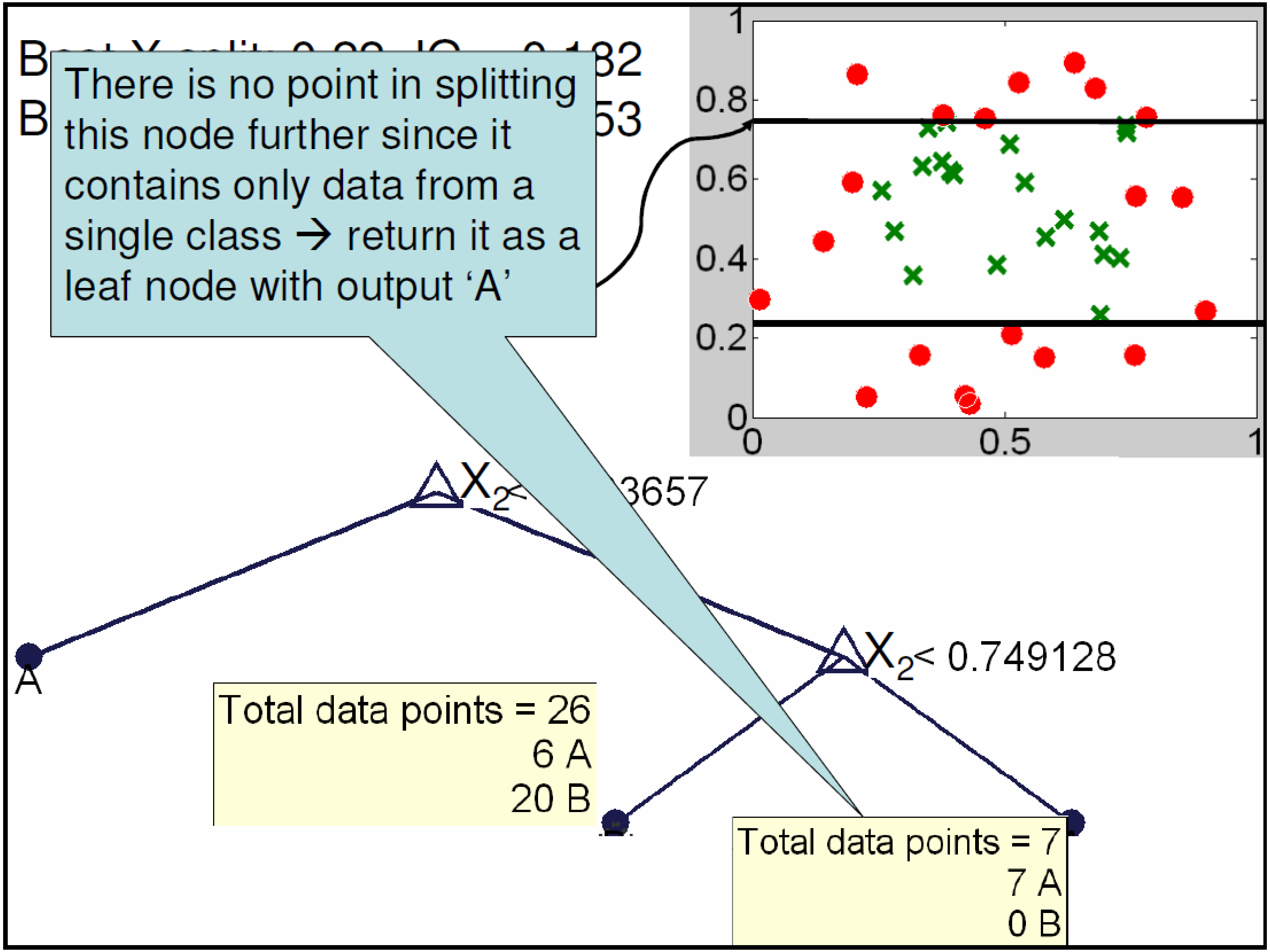
Best X_1 split: 0.22, IG = 0.182

Best X_2 split: 0.75, IG = 0.353

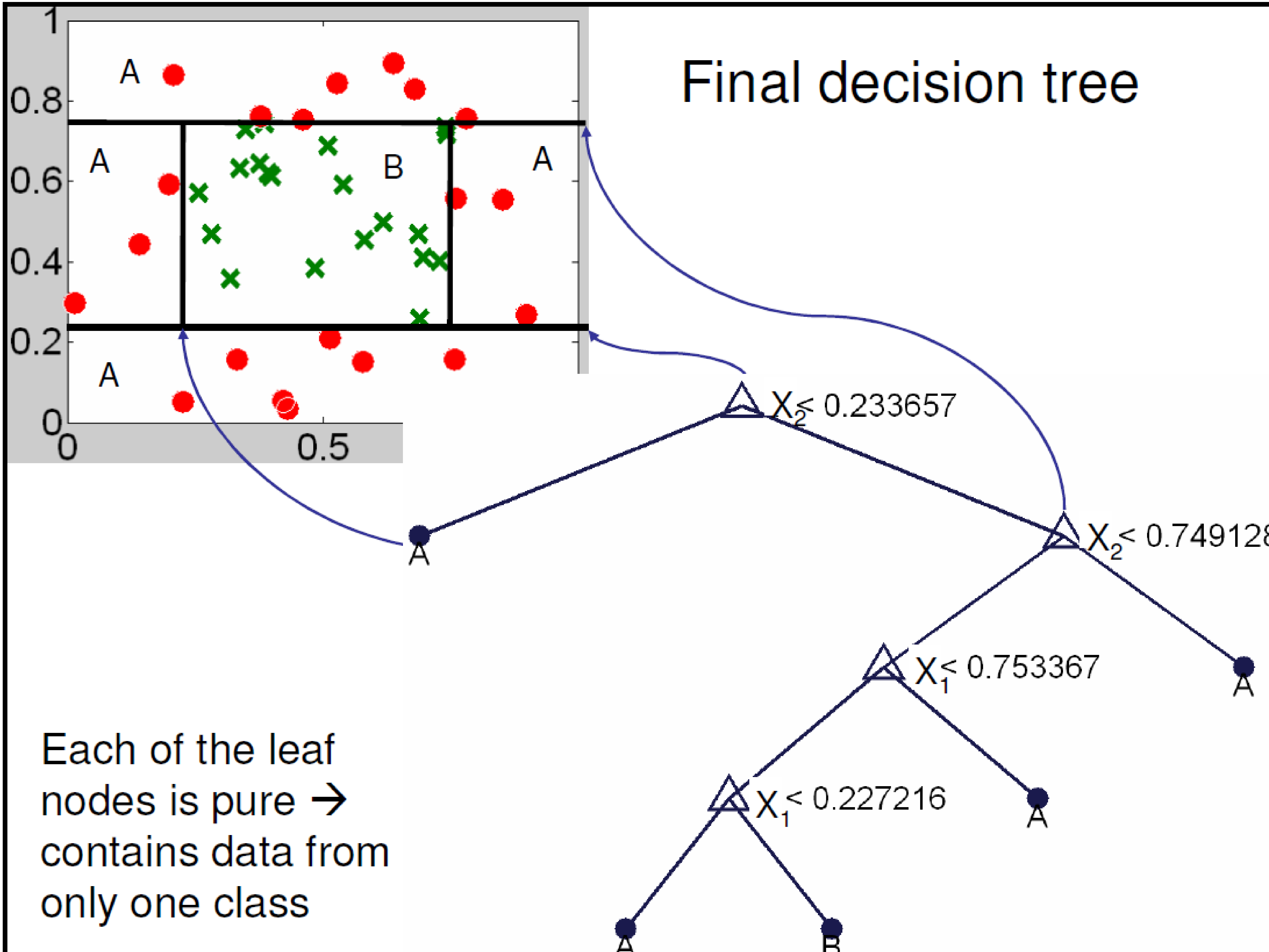
Split on X_2 with 0.75






There is no point in splitting this node further since it contains only data from a single class → return it as a leaf node with output 'A'
X2< 0.749128
A
Total data points = 26
6 A
20 B
Total data points = 7
7 A
0 B
0
0.2
0.4
0.6
0.8
1
0
0.5
1

Final decision tree

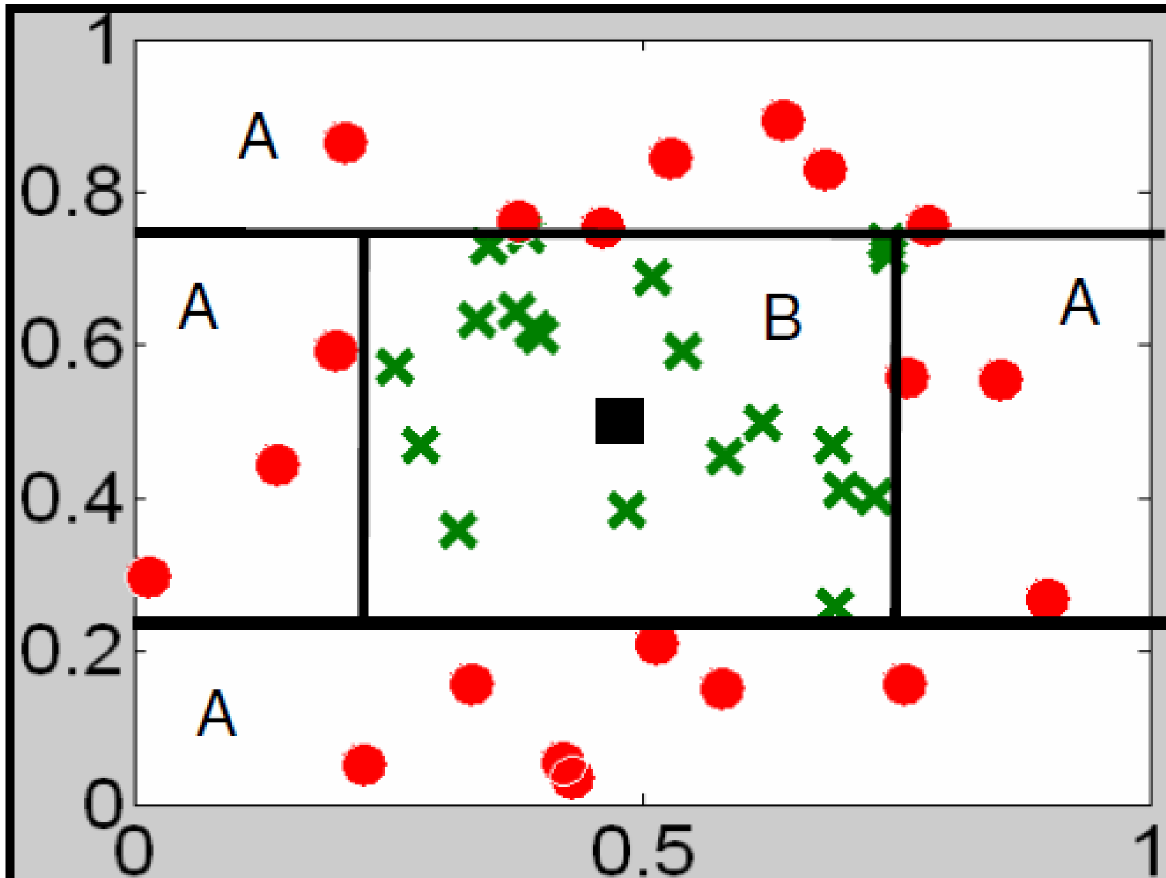


Each of the leaf nodes is pure → contains data from only one class

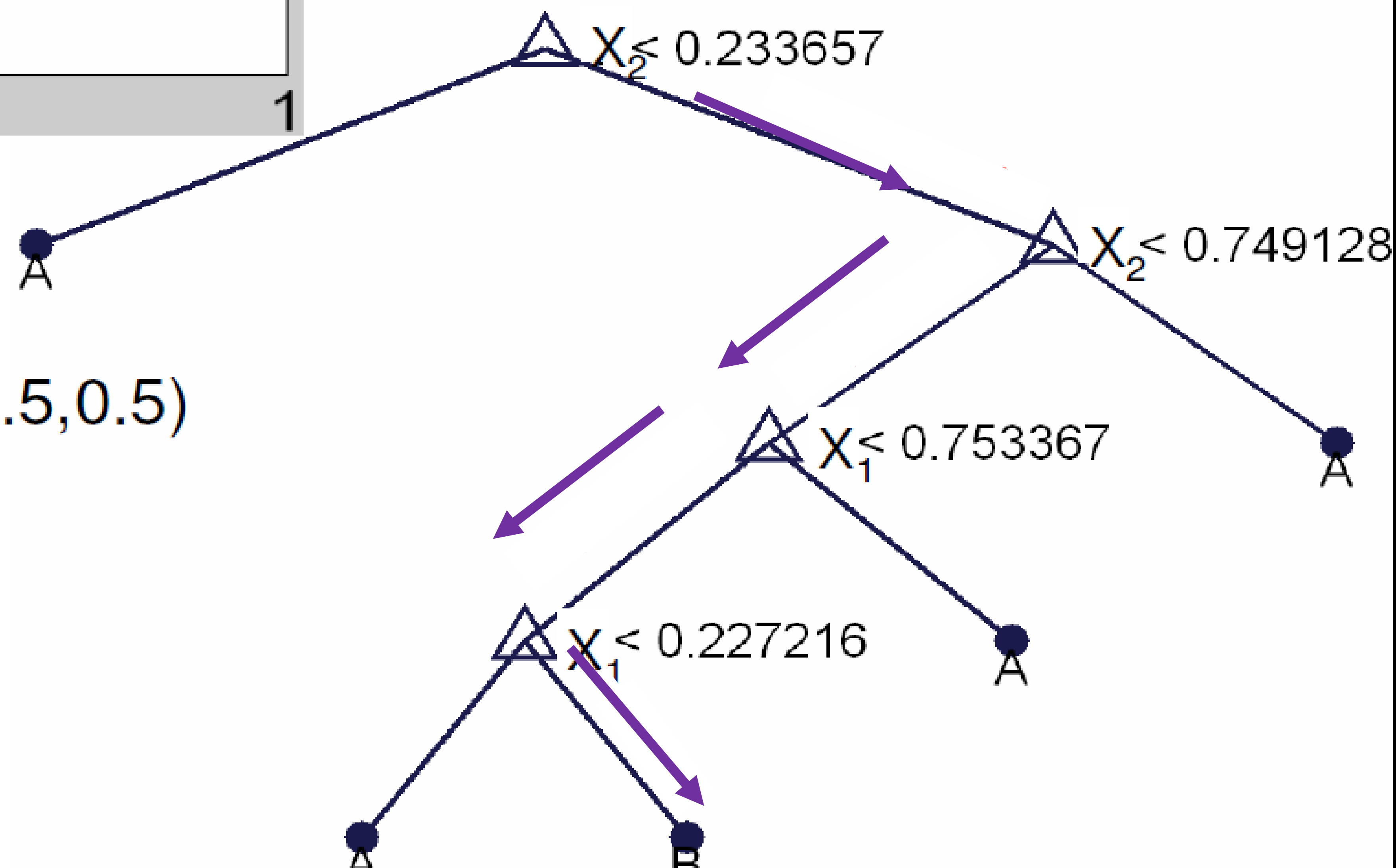
Final decision tree

Given an input (X,Y) → Follow the tree down to a leaf.

Return corresponding output class for this leaf



Example (X,Y) = (0.5,0.5)

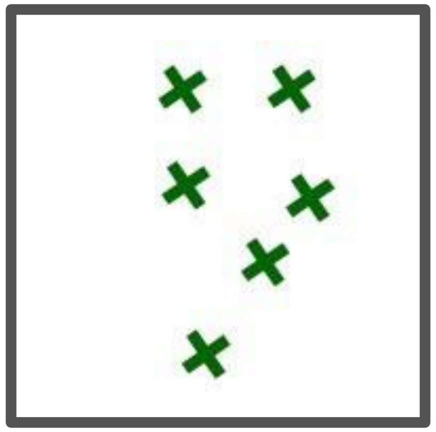


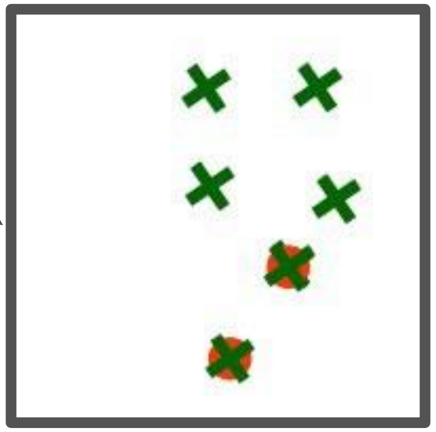
Basic Questions

- How to choose the attribute/value to split on at each level of the tree?
- When to stop splitting? When should a node be declared a leaf?
- If a leaf node is impure, how should the class label be assigned?
- If the tree is too large, how can it be pruned?

When to stop splitting? Common strategies:

1. Pure and impure leave nodes
   - All points belong to the same class; OR
   - All points from one class completely overlap with points from another class (i.e., same attributes)
     - Output majority class as this leaf's label





2. Node contains points fewer than some threshold
3. Node purity is higher than some threshold
4. Further splits provide no improvement in *training loss*
   ($loss(T) <= loss(T_L) + loss(T_R)$)

Graphics from: http://www.cs.cmu.edu/afs/cs.cmu.edu/academic/class/15381-s06/www/DTs.pdf

Decision Tree Algorithm (Continuous Attributes)

- LearnTree(X, Y)
  - Input:
    - Set X of R training vectors, each containing the values $(x_1,..,x_M)$ of M attributes $(X_1,..,X_M)$
    - A vector Y of R elements, where y_j = class of the j[th] datapoint
  - If all the datapoints in X have the same class value y
    - Return a leaf node that predicts y as output
  - If all the datapoints in X have the same attribute value $(x_1,..,x_M)$
    - Return a leaf node that predicts the majority of the class values in Y as output
  - Try all the possible attributes X_j and threshold t and choose the one, j^*, for which IG($Y|X_j,t$) is maximum
  - X_L, Y_L = set of datapoints for which $x_{j^*} < t$ and corresponding classes
  - X_H, Y_H = set of datapoints for which $x_{j^*} >= t$ and corresponding classes
  - Left Child ← LearnTree(X_L, Y_L)
  - Right Child ← LearnTree(X_H, Y_H)

Decision Tree Algorithm (Discrete Attributes)

- LearnTree(X, Y)
  - Input:
    - Set X of R training vectors, each containing the values $(x_1,..,x_M)$ of M attributes $(X_1,..,X_M)$
    - A vector Y of R elements, where y_j = class of the jth datapoint
  - If all the datapoints in X have the same class value y
    - Return a leaf node that predicts y as output
  - If all the datapoints in X have the same attribute value $(x_1,..,x_M)$
    - Return a leaf node that predicts the majority of the class values in Y as output
  - Try all the possible attributes X_j and choose the one, j^*, for which IG($Y|X_j$) is maximum
  - For every possible value v of X_{j^*}:
    - X_v, Y_v = set of datapoints for which $x_{j^*} = v$ and corresponding classes
    - Child$_v$ ← LearnTree(X_v, Y_v)

Decision Trees So Far

- Given N observations from training data, each with D attributes X and a class attribute Y, construct a sequence of tests (decision tree) to predict the class attribute Y from the attributes X
- Basic strategy for defining the tests ("when to split") → maximize the information gain on the training data set at each node of the tree
- Problems (next):
  - Computational issues → How expensive is it to compute the IG
  - The tree will end up being much too big → *pruning*
  - Evaluating the tree on training data is dangerous → *overfitting*

Basic Questions

- How to choose the attribute/value to split on at each level of the tree?
- When to stop splitting? When should a node be declared a leaf?
- If a leaf node is impure, how should the class label be assigned?
- If the tree is too large, how can it be pruned?

What will happen if a tree is too large?

Overfitting

High variance

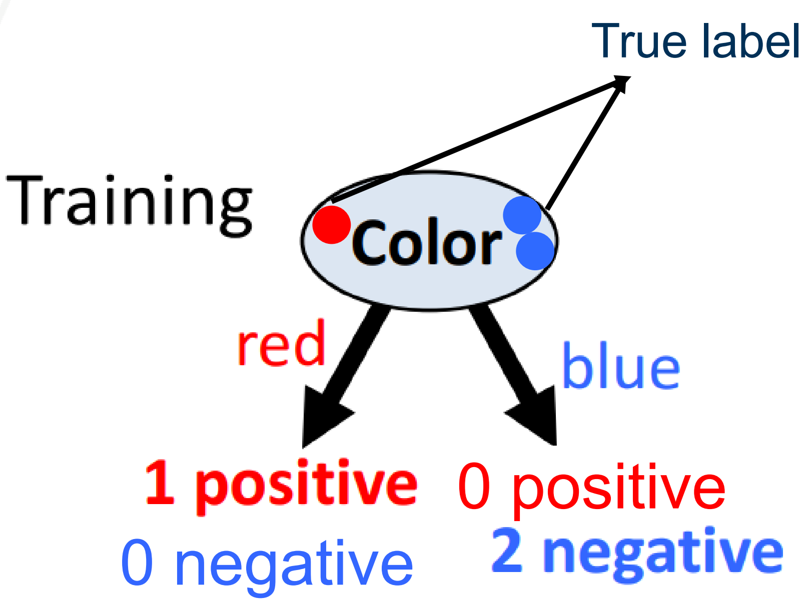
Instability in predicting test data

How to avoid overfitting?

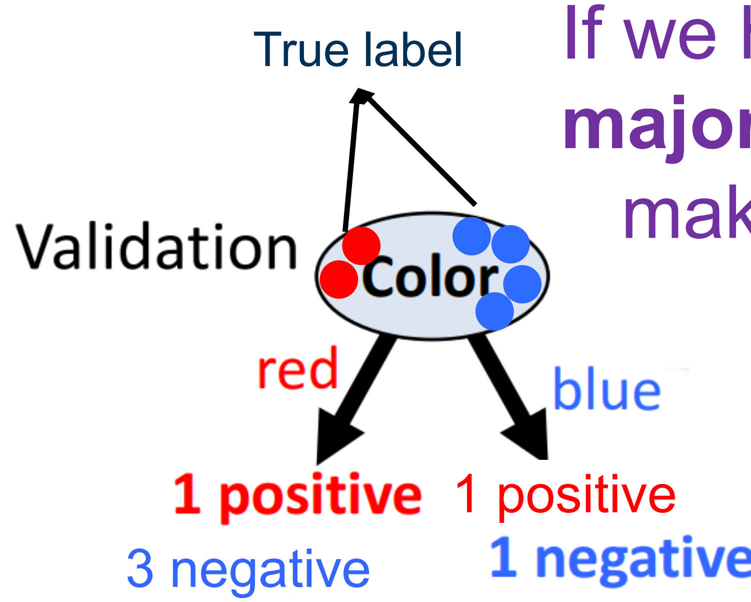
- Acquire more training data
- Remove irrelevant attributes (manual process – not always possible)
- Grow full tree, then post-prune
- Ensemble learning

How to decide to remove it a node using pruning

- Pruning of the decision tree is done by replacing a whole subtree by a leaf node.
- The replacement takes place if a decision rule establishes that the expected error rate in the subtree is greater than in the single leaf.



3 training data points
Actual Label: 1 positive and 2 negative
Predicted Label: 1 positive and 2 negative
3 correct and **0** incorrect



6 validation data points
Actual label:2 positive and 4 negative
Predicted Label: 4 positive and 2 negative
2 correct and **4** incorrect

If we had simply predicted the **majority class** (negative), we make 2 errors instead of 4

Pruned!

Reduced-Error Pruning

Split data into training and validation sets

Grow tree based on training set

Do until further pruning is harmful:

1. Evaluate impact on validation set of pruning each possible node (plus those below it)

2. Greedily remove the node that most improves validation set accuracy